Fundamentals of Fog and Edge Computing

BCSE313L

WINTER 2024 - 2025

Module 3 - Orchestration of Network Slices in Fog, Edge, and Clouds

Introduction

- Digital transformation is reshaping various sectors through innovative applications such as smart cities, vehicle-to-vehicle (V2V) communication, virtual reality (VR), augmented reality (AR), and remote medical surgery.
- These applications have **distinct connectivity** and **performance requirements**, presenting significant challenges for traditional network infrastructures that rely on a one-size-fits-all approach.

Digital Transformation Has Led Explosion in

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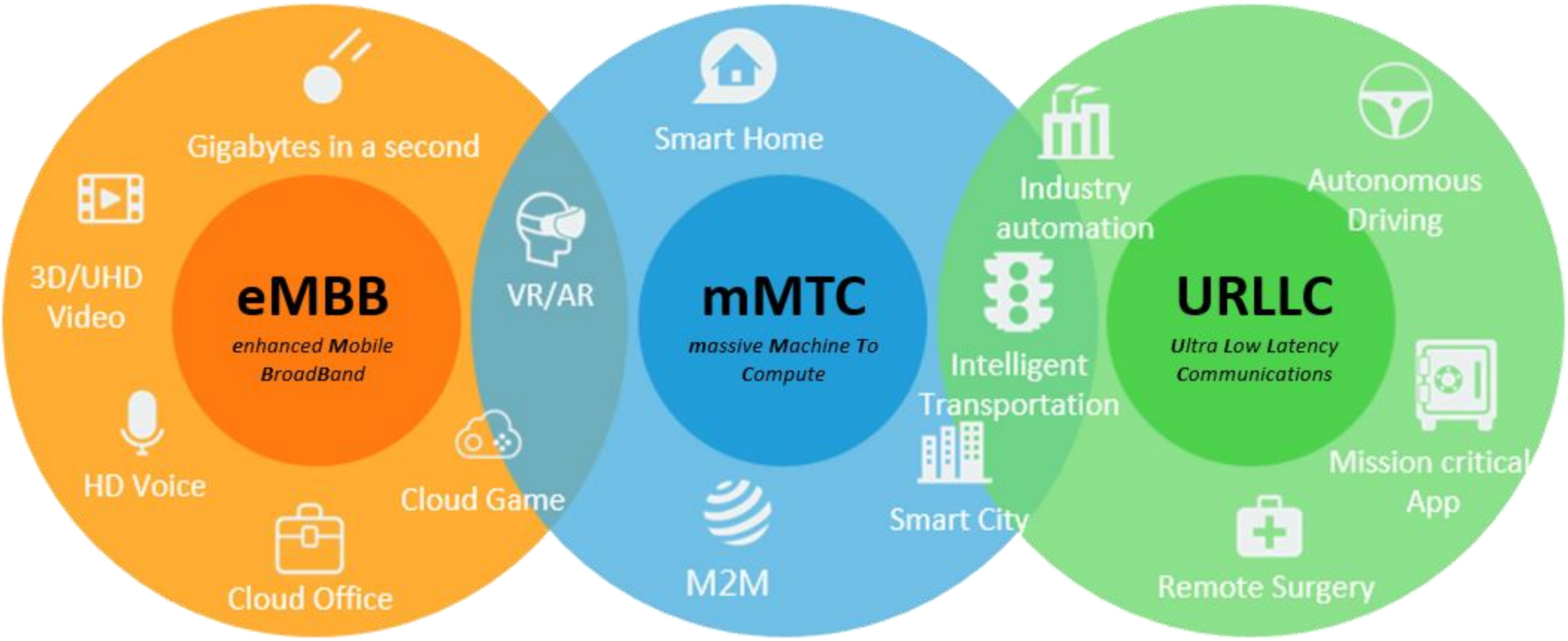
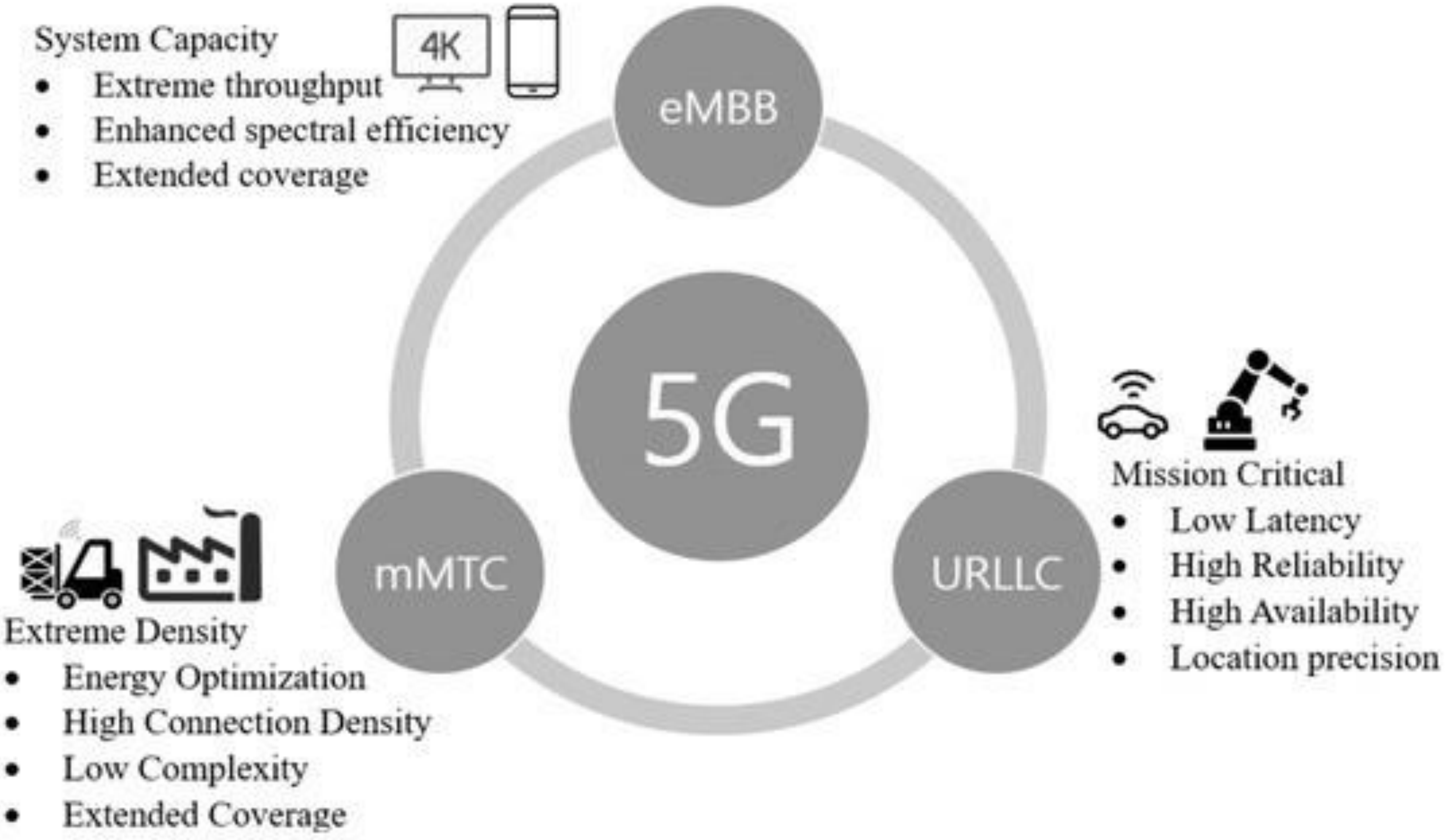


Introduction - Key Use Cases and Quality of Service Requirements

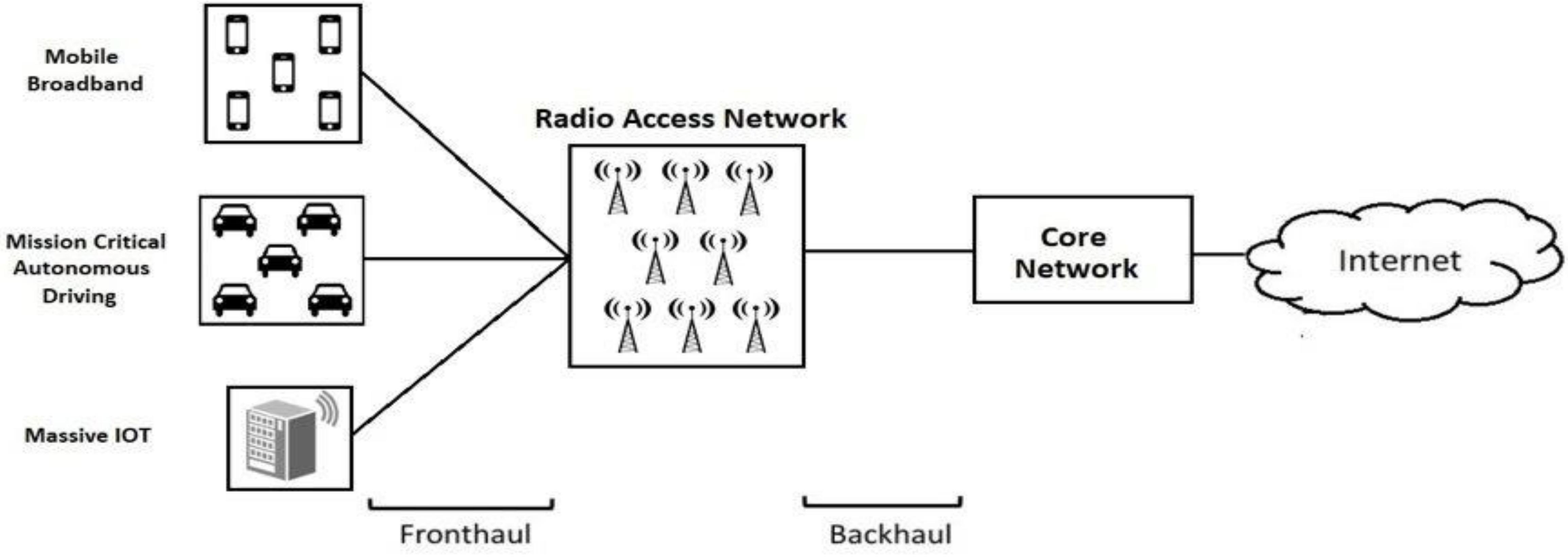


Introduction

- The 5G infrastructure public-private partnership (5G-PPP) has categorized these applications into three main use case families:
  - Enhanced mobile broadband (eMBB),
  - Massive machine-type communications (mMTC)
  - Ultra-reliable low-latency communication (uRLLC)

Introduction – Traditional Network Architecture Limitations

- The diverse quality of service (QoS) requirements of these use cases highlight the limitations of traditional network architectures, which cannot efficiently support the varying demands of different vertical services.



Introduction – Traditional Network Architecture Limitations

- Manual Configurations
- Limited Scalability
- Inconsistent Quality of Service (QoS)
- Complexity in Management
- Rigid Architecture
- Vendor Dependence

Introduction – Traditional Network Architecture Limitations

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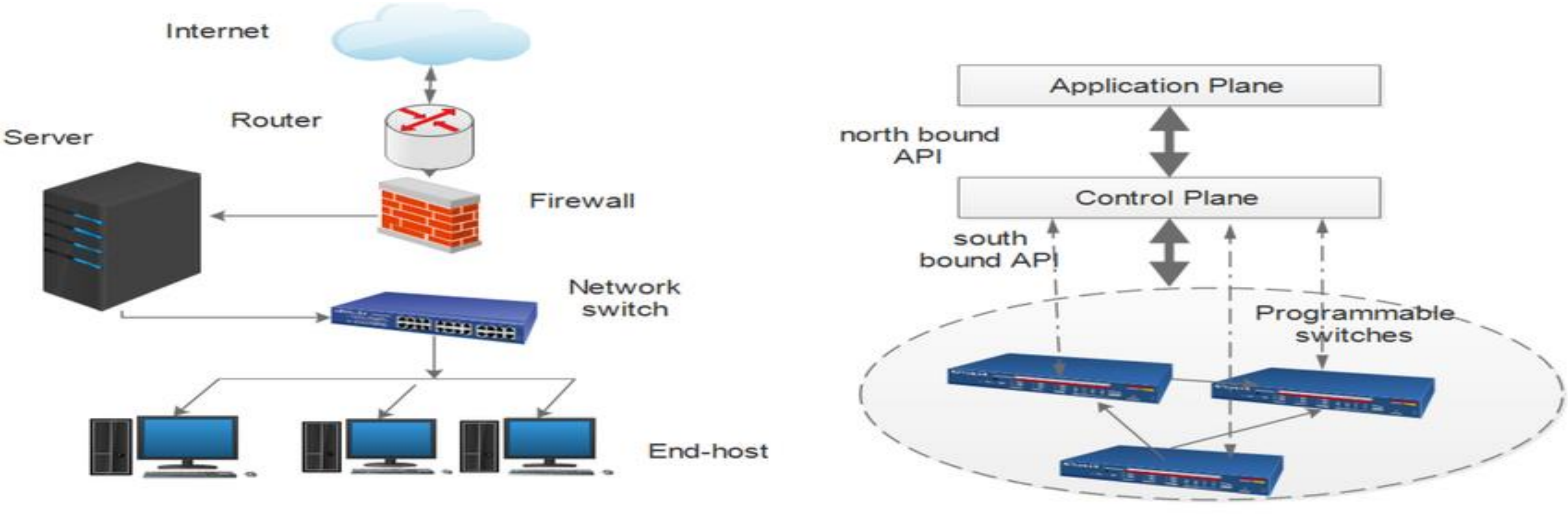


a Traditional Network architecture

b SDN architecture

Introduction – Traditional Network Architecture Limitations

Manual Configuration:

- Traditional networks rely on manual configuration of individual devices such as routers and switches. This process is time-consuming, increases operational complexity, and is prone to human error, making it difficult to maintain consistency across the network

Limited Scalability:

- These architectures struggle to scale effectively with the increasing number of devices and bandwidth demands. They often require significant reconfiguration when new devices are added, which can lead to oversubscription and performance issues

Introduction – Traditional Network Architecture Limitations

Inconsistent Quality of Service (QoS):

Implementing consistent QoS policies across numerous devices is challenging due to the manual nature of configurations. This inconsistency can lead to inadequate performance for applications requiring high reliability and low latency, such as uRLLC

Complexity in Management

The complexity of managing traditional networks increases with the number of protocols and devices involved. This complexity makes it difficult to troubleshoot issues that affect multiple devices, leading to longer resolution times and increased downtime

Introduction – Traditional Network Architecture Limitations

Rigid Architecture:

Traditional networks have a hierarchical architecture that is not easily adaptable to changing business needs or traffic patterns. This rigidity prevents dynamic resource allocation, which is essential for accommodating varying demands from different applications

Vendor Dependence:

Many traditional networks are built on proprietary hardware and software solutions, which can limit flexibility and increase costs when integrating new technologies or scaling services

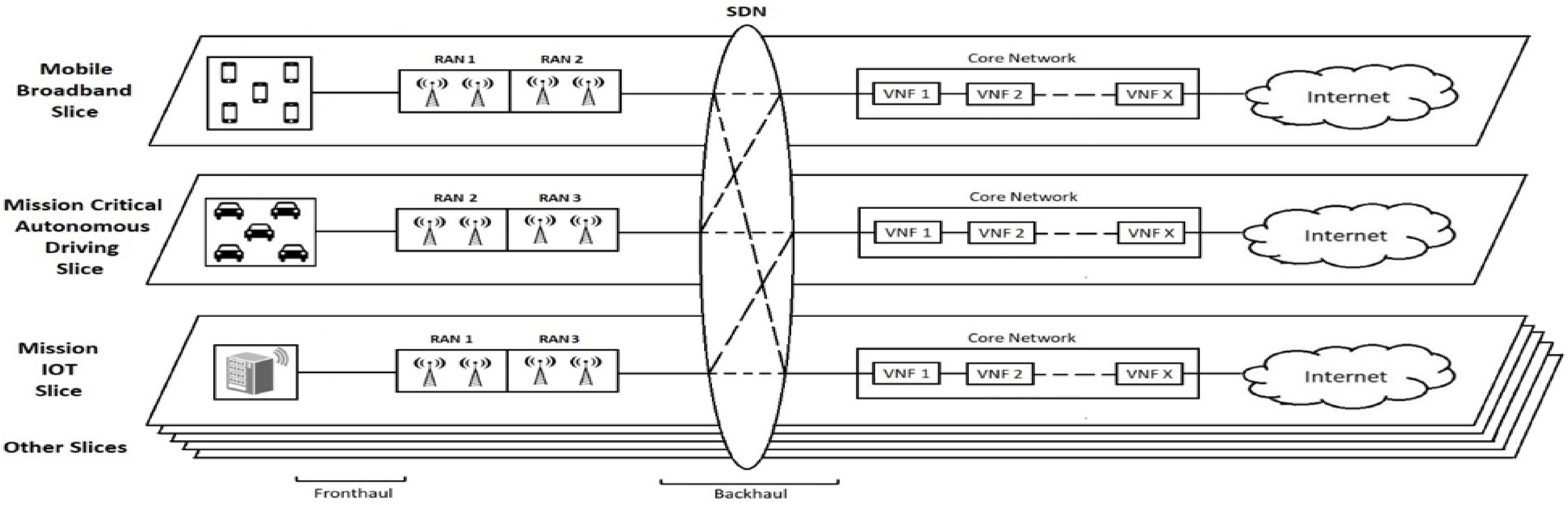
Network Slicing - as a Solution

- A promising approach to address these challenges is network slicing, which partitions a physical network into multiple isolated logical networks.

- This concept mirrors server virtualization in cloud computing, allowing for dynamic resource allocation based on specific application needs.

Network Slicing - as a Solution

- Network slicing enables the creation of tailored virtual networks that can adapt to the unique requirements of different services, effectively providing a network-as-a-service (NaaS) model



Network Slicing - Benefits

Cost Efficiency

- Reduces the need for dedicated physical infrastructure by allowing multiple services to share the same resources

Flexibility

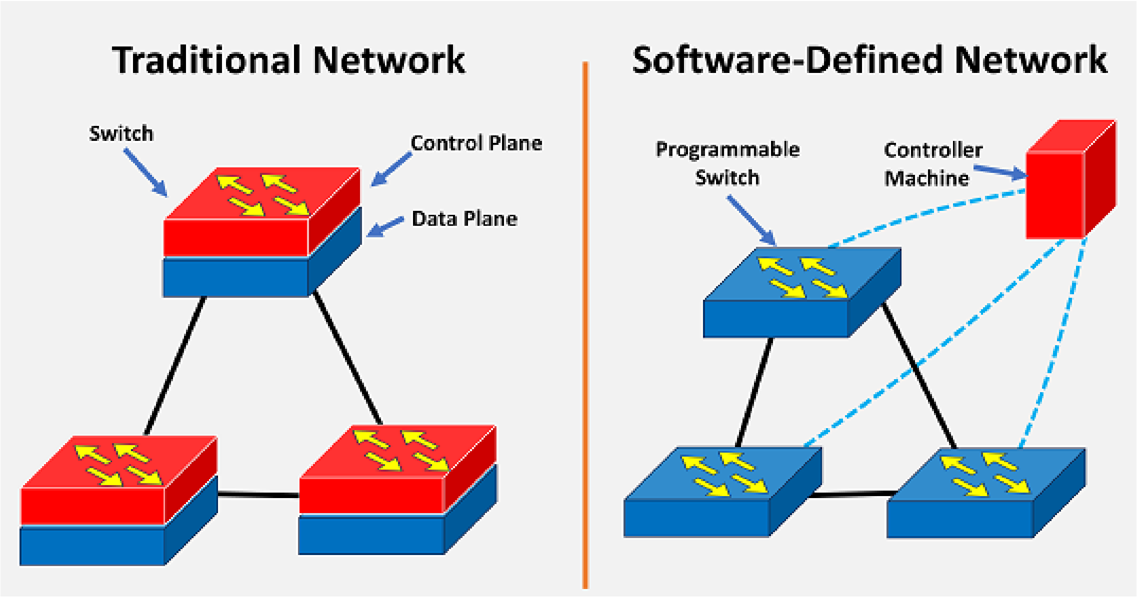
- Operators can quickly deploy and manage network slices tailored to specific customer needs.

Scalability

- Easily scales resources up or down based on demand, accommodating varying service requirements

Role of SDN and NFV in Network Slicing

- Software-Defined Networking (SDN) and Network Function Virtualization (NFV)/Network virtualized function (NVF) are foundational technologies that enhance network slicing capabilities:
  - SDN separates control and data planes, providing centralized management and improved visibility across the network.
  - NFV shifts network functions from hardware to software applications on general-purpose hardware, increasing elasticity and reducing costs



Network Slicing in 5G

- Network slicing in 5G refers to the **creation of multiple virtual networks on top of a single physical network**.

- Each slice operates independently with its own resources, topology, traffic flow, management policies, and protocols, allowing for tailored connectivity for specific applications or services

Network Slicing in 5G

- The primary goal of network slicing is to meet the **unique requirements of various use cases**—ranging from enhanced mobile broadband (eMBB) for high-bandwidth applications to ultra-reliable low-latency communication (uRLLC) for critical services like remote surgery

Network Slicing in 5G - Challenges

Resource Provisioning

- Managing resource allocation among multiple slices is complex due to varying resource affinities that can change over time

Mobility Management

- Handling mobility across slices can complicate the management of resources and connectivity

End-to-End Orchestration

- Coordinating the orchestration and management of slices across different domains (like RAN and core networks) adds complexity to implementation

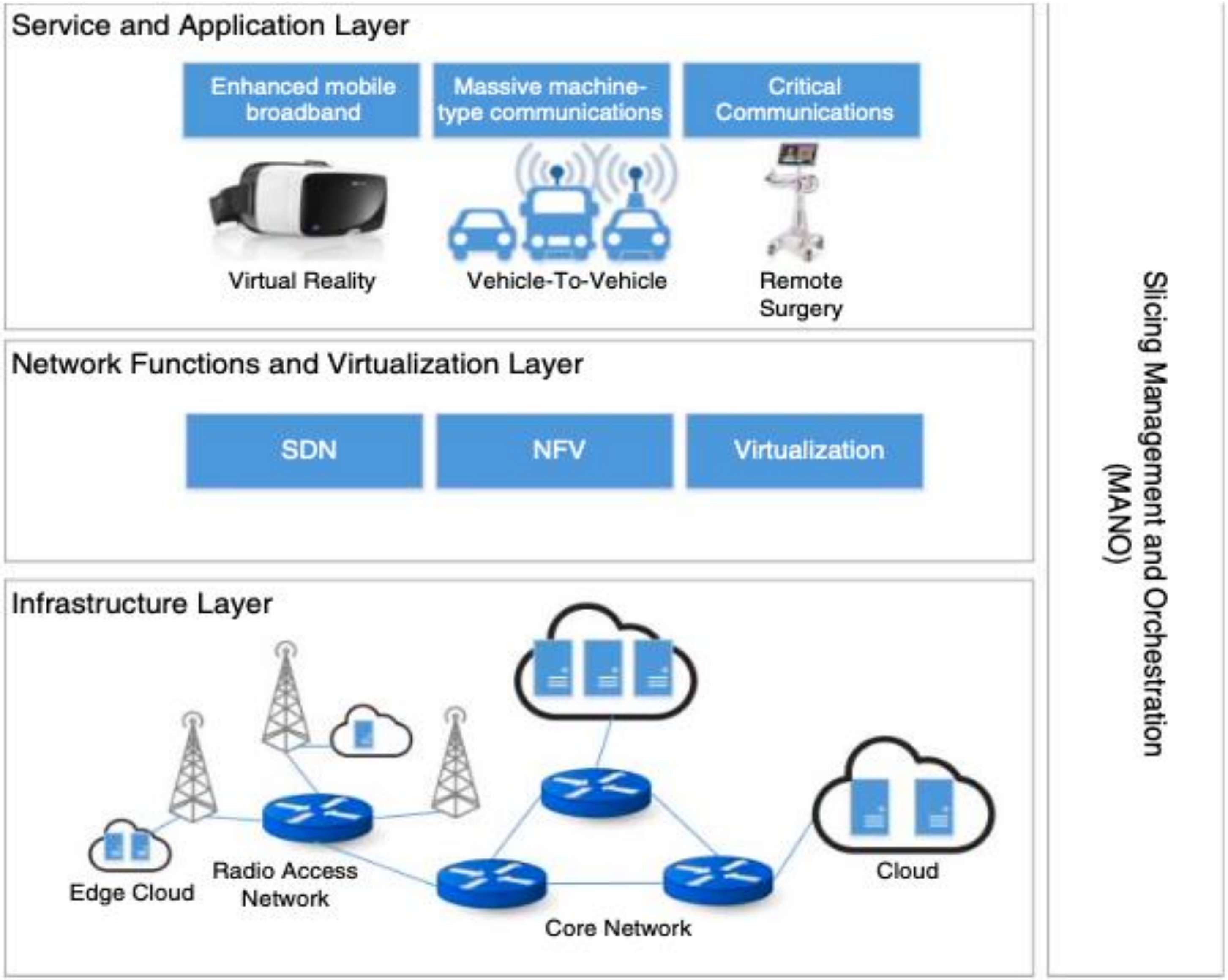
Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**: This layer includes the physical resources such as servers, storage, and networking equipment.

- **Network Function Layer:** This layer encompasses the virtualized network functions that can be deployed within each slice.

- **Service Layer**: This layer focuses on the specific services offered to end-users through the various slices

Network Slicing in 5G – Generic Framework



Network Slicing in 5G – Generic Framework

Use Case
(Connected Devices, Applications, Services, Specific Utility Expectations)

Mapping Functions
(SLAs, Mapping N/w Functions to NVF to Physical resource)

Service and Application Layer

Lifecycle Management
(Managing NVFs, Optimal Placements of NVFs in physical resources)

Chaining of Slices
(Make Slices interact, leverage combined functionalities as needed)

Technical Aspects
(Coarse grained and Fine grained NVFs managing, SDN implementation)

Network Function and Virtualization Layer

Physical Network Architecture
(Remote Cloud, Edge Cloud, Radio Access Network, Cloud-RAN)

Resource Abstraction
(Software Defined Technique, Virtual Storage, Network, Compute, Dynamic allocation)

Policy Implementation
(Policies for deploy, control, manage, Orchestrate for allocation)

Resource Allocation
(Allocation of abstracted resources to the network slices)

Infrastructure Layer

Create Virtual instances from Physical Resource
Establishment of Network Slices

Maps N/W Functions to Virtualized Instance
Construct Service Chains
Allocation of Resources & Functions

Mange Communication between Layers
Lifecycle Management
Adapting and Scaling to needs

Slicing Management and Orchestration (MANO)

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - It defines the actual physical network topology that supports multiple virtual networks.
  - *Physical Network Architecture*: The infrastructure layer extends from edge cloud environments to remote cloud setups, encompassing both the radio access network (RAN) and the core network. This layered approach allows for efficient resource allocation and management across various network components .

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - ***Resource Abstraction***: Utilizing software-defined techniques, this layer facilitates resource abstraction, allowing for dynamic allocation of compute, storage, bandwidth, and other resources to different network slices. This abstraction is critical for enabling flexibility and responsiveness to varying service demands .

Network Slicing in 5G – Generic Framework

- **Infrastructure Layer**
  - ***Policy Implementation***: Within the infrastructure layer, various policies are established to deploy, control, manage, and orchestrate the underlying physical infrastructure. These policies ensure that resources are allocated effectively to meet the specific needs of each slice while maintaining overall network performance .
  - **End-to-End Resource Allocation**: The infrastructure layer plays a crucial role in allocating resources to network slices in a manner that supports upper layers. This allocation is done contextually, ensuring that each slice can operate independently with the necessary resources to fulfill its requirements

Network Slicing in 5G – Generic Framework

- **Network Function and Virtualization Layer**
  - The Network Function and Virtualization Layer operates above the infrastructure layer and is responsible for managing the lifecycle of virtual resources and network functions
  - ***Lifecycle Management***: This layer executes all necessary operations for managing virtualized network functions (VNFs), ensuring optimal placement of these functions within the available physical resources.

Network Slicing in 5G – Generic Framework

- **Network Function and Virtualization Layer**
  - ***Chaining of Slices***: It facilitates the chaining of multiple slices to meet specific service requirements. By coordinating how different slices interact, this layer ensures that applications can leverage combined functionalities across slices .
  - ***Technical Aspects***: Key technologies such as Software-Defined Networking (SDN) and Network Function Virtualization (NFV) are integral to this layer. They enable efficient management of both coarse-grained and fine-grained network functions across the core and local RAN .

Network Slicing in 5G – Generic Framework

- **Service and Application Layer**
  - The Service and Application Layer represents the topmost layer where various applications and services operate:
  - ***Use Cases***: This layer includes connected devices such as vehicles, IoT devices, mobile phones, and VR appliances. Each application or service has specific utility expectations from the underlying networking infrastructure.
  - ***Mapping Functions***: Based on high-level service requirements, virtualized network functions are mapped to physical resources to ensure compliance with Service Level Agreements (SLAs). This mapping ensures that each application receives the necessary resources without violating performance guarantees .

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - This layer oversees the functionality of the underlying infrastructure, network functions, and service layers by performing three main tasks:

  - ***Creation of Virtual Network Instances:*** The MANO layer is responsible for creating virtual network instances based on the physical network resources defined in the infrastructure layer. This process allows for the establishment of independent slices that can cater to various applications and services.

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - ***Mapping Network Functions:*** It maps network functions to these virtualized network instances, facilitating the construction of service chains. This mapping is essential for ensuring that specific service requirements are met through the appropriate allocation of resources and functionalities from the network function and virtualization layer.

Network Slicing in 5G – Generic Framework

- **Slicing Management and Orchestration (MANO) in 5G**
  - *Lifecycle Management:* The MANO layer maintains communication between the service/application layer and the network slicing framework. This function involves managing the lifecycle of virtual network instances and dynamically adapting or scaling virtualized resources in response to changing contexts or demands

Network Slicing in 5G – Generic Framework

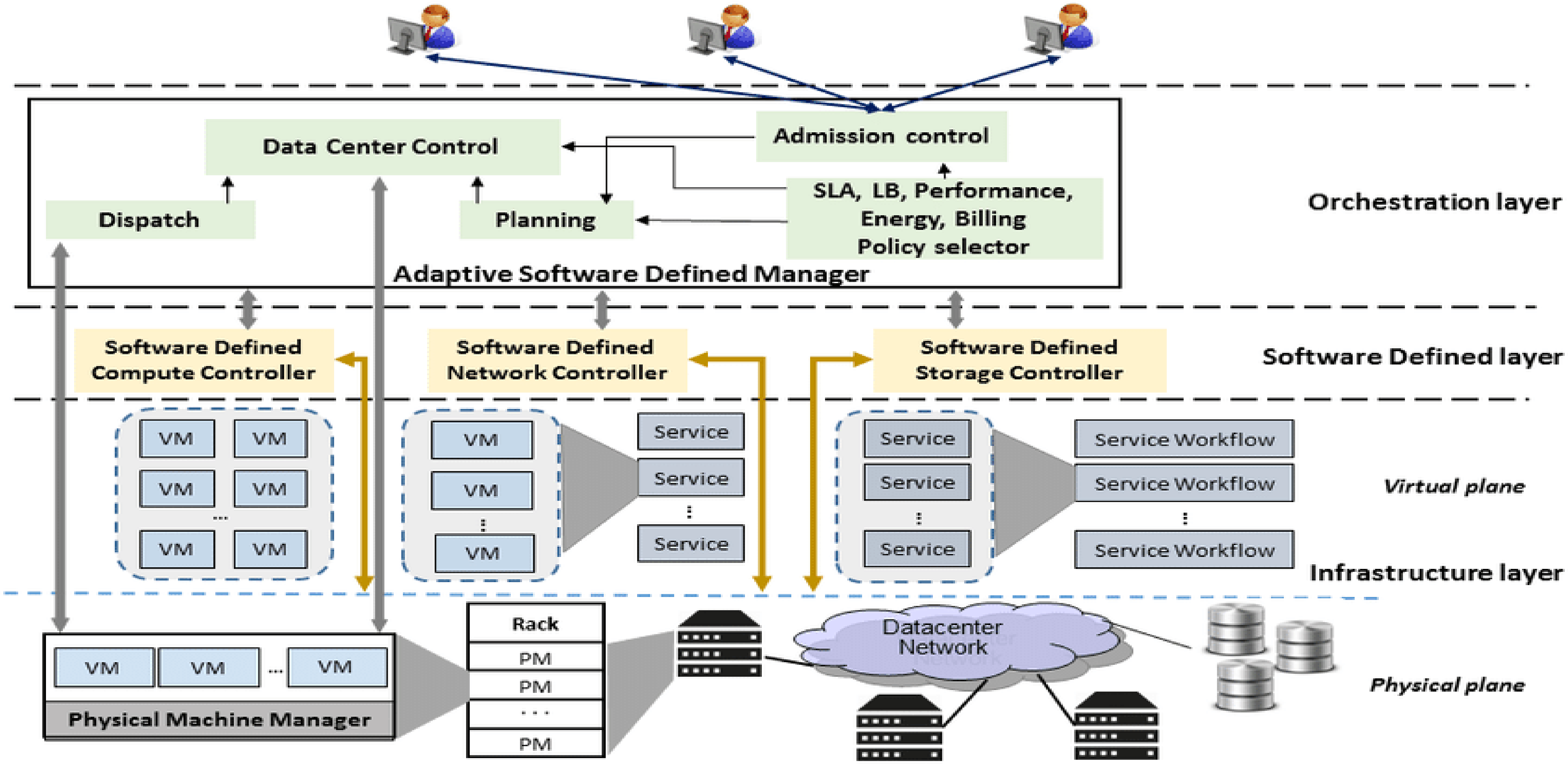
Characteristics of Cloud-Native Network Architecture for 5G	**Cloud Data Center-Based Architecture:**	It provides a logically independent network slicing capability over a cloud data center infrastructure, supporting various application scenarios.
	Cloud-RAN Implementation:	The architecture employs Cloud-RAN to build radio access networks (RAN), facilitating many connections and enabling on-demand deployments of RAN functions as required by 5G.
	Simplified Core Network Architecture:	The core network design is streamlined, allowing for on-demand configuration of network functions through user and control plane separation, unified database management, and component-based functionalities.
	Automated Network Slicing Services:	The architecture implements network slicing services automatically to reduce operational expenses, enhancing efficiency in resource utilization .

Network Slicing in Software Defined Clouds

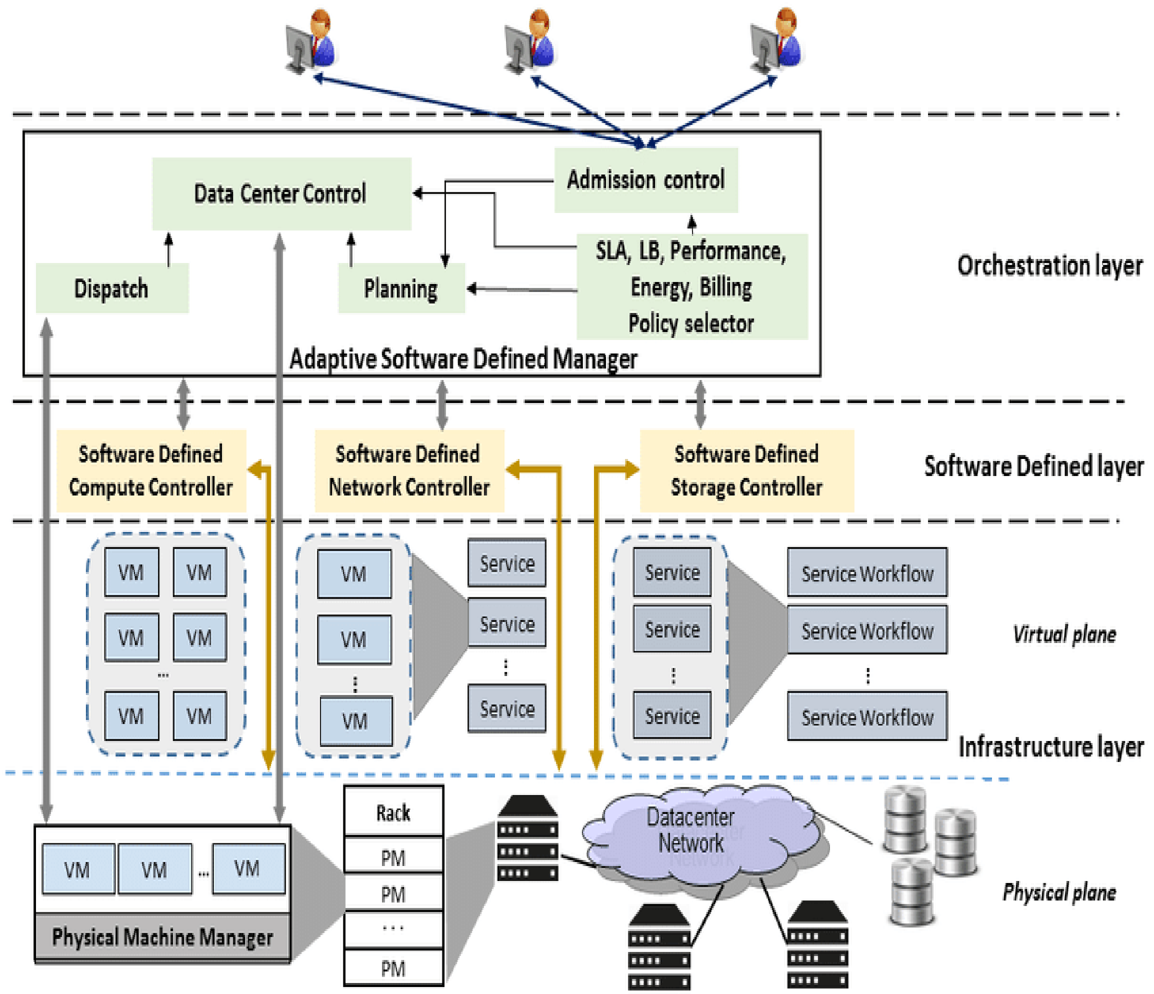
- Virtualization technology has been crucial for resource management and optimization in cloud data centers for the last decade.

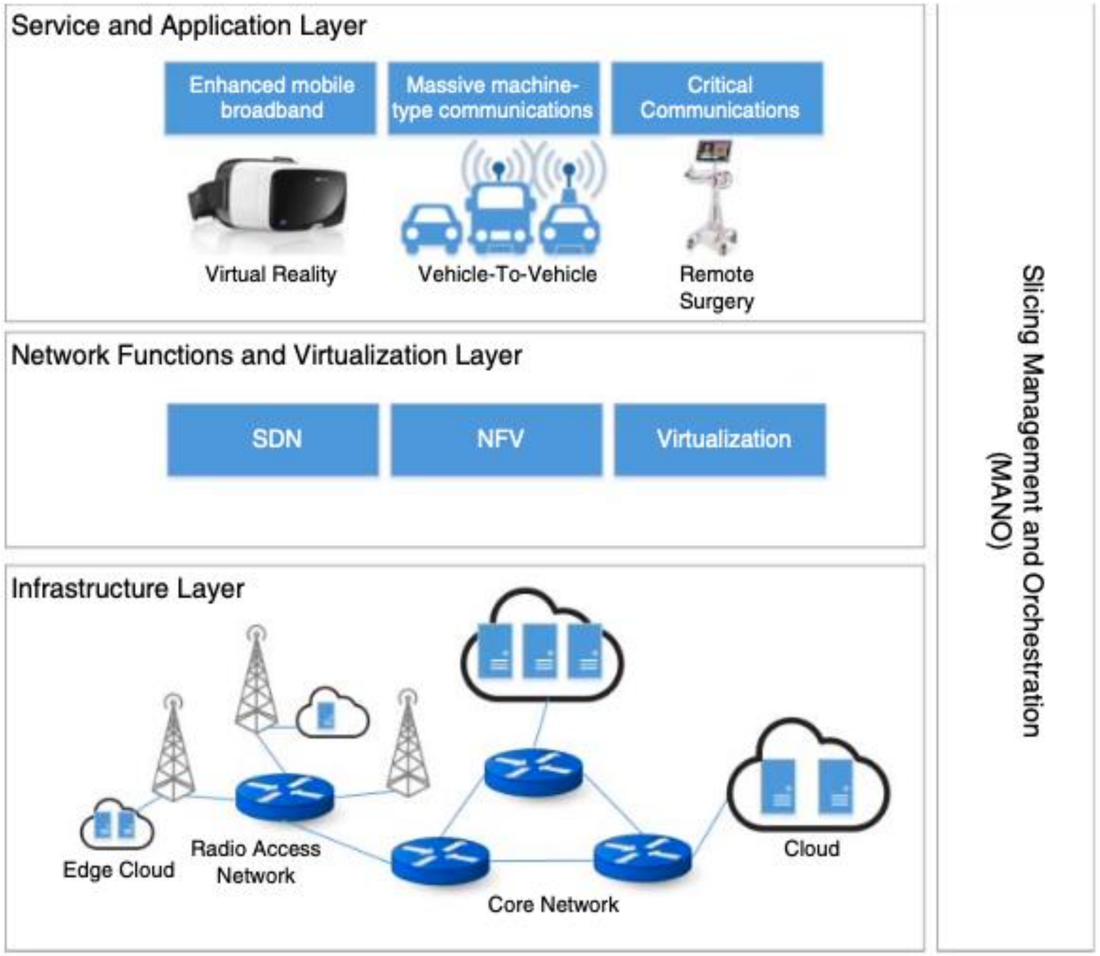
- Research has focused on VM placement and virtual machine (VM) migration to improve utilization and efficiency of physical and virtual servers.

Software Defined Clouds Architecture



Software Defined Clouds Architecture

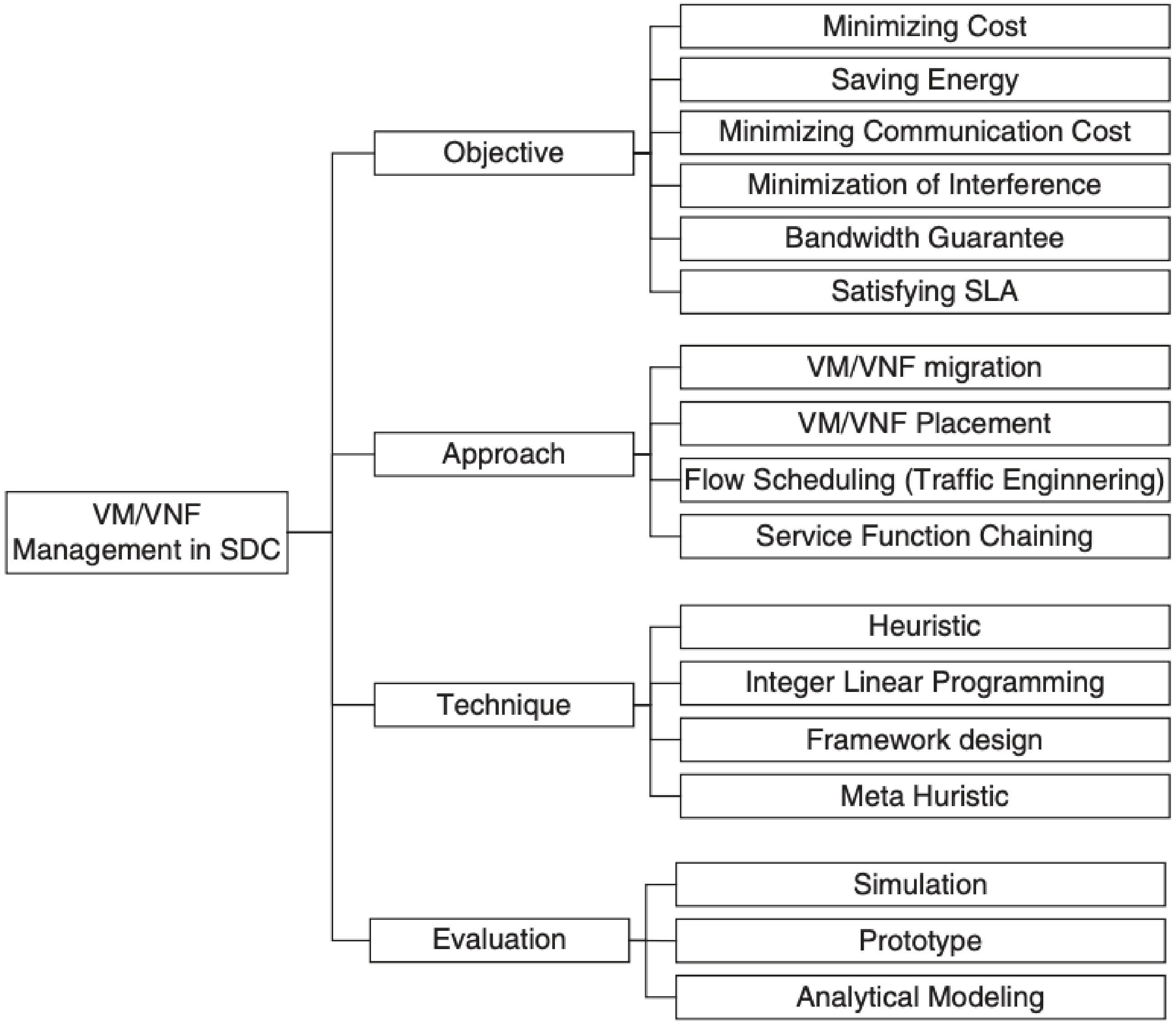




Perspectives on Network Slicing

- It refers to different viewpoints or approaches to understanding and managing network slicing in cloud data centers.
  - **Network-aware VM management:** This perspective focuses on managing virtual machines (VMs) in a way that takes into account the network resources and requirements.
  - **Network-aware VM migration**: This perspective focuses on migrating VMs between different physical or virtual servers while considering network constraints and optimization.
  - **VNF management**: This perspective focuses on managing Virtual Network Functions (VNFs), which are software implementations of network functions, such as firewalls or routers.

Network-Aware VM/VNF Management Taxonomy



Network-Aware VM/VNF Management Taxonomy

Project	Objectives	Approach/Technique	Evaluation
Cziva et al. [29]	Minimization of the network communication cost	VM migration – Framework Design	Prototype
Wang et al. [30]	Reducing the number of hops between communicating VMs and network power consumption	VM placement – Heuristic	Simulation
Remedy [31]	Removing congestion in the network	VM migration – Framework Design	Simulation
Jiang et al. [32]	Minimization of the network communication cost	VM Placement and Migration – Heuristic (Markov approximation)	Simulation
VMPlanner [33]	Reducing network power consumption	VM placement and traffic flow routing - Heuristic	Simulation
PLAN [35]	Minimization of the network communication cost while meeting network policy requirements	VM Placement - Heuristic	Prototype/ Simulation

Network-Aware Virtual Machine Migration Planning

Project	Objectives	Approach/Technique	Evaluation
Bari et al. [37]	Finding sequence of migrations while migration time is minimized	VM migration – Heuristic	Simulation
Ghorbani et al. [38]	Finding sequence of migrations while imposing bandwidth guarantees	VM migration – Heuristic	Simulation
Li et al. [39]	Finding sequence of migrations and destination hosts to balance the load	VM migration – Heuristic	Simulation
iAware [40]	Minimization of migration and co-location interference among VMs	VM migration – Heuristic	Prototype/ Simulation

Virtual Network Functions Management

Project	Objectives	Approach/Technique
VNF-P	Handling burst in network services demand while minimizing the number of servers	Resource allocation - Integer linear programming (ILP)
Cloud4NFV	Providing network function as a service	Service provisioning – Framework design
vConductor	Virtual network services provisioning and management	Service provisioning – Framework design
MORSA	Multi objective placement of virtual services	Placement - Multi-objective genetic algorithm
TVM	Reducing number of hops in service chain	VNF migration - heuristic
SOVWin	Increasing user requests acceptance rate and minimization of SLA violation	VNF placement - heuristic
Clayman et al.	Providing automatic placement of the virtual nodes	VNF placement - heuristic
T-NOVA	Building a marketplace for VNF	Marketplace – framework design
UNIFY	Automated, dynamic service creation and service function chaining	Service provisioning– framework design

Network Slicing Management in Edge and Fog

- Fog computing extends cloud computing to the edge of the network, providing real-time and low-latency processing.

- **Key challenges**: Integrating fog/edge computing with SDN/NFV for efficient management and orchestration of network services.

Network Slicing Management in Edge and Fog

Create SDN Controllers	• Network Slices can be defined and managed programmatically • dynamic resource allocation and reconfiguration
Apply and Use NFV	• Virtualization of network functions • Virtualised Network functions can be deployed and scaled on demand within network slices
Use intelligent orchestration and automation tools.	• Automate the provisioning, configuration, and management of network slices
FOSA or Platforms or Frameworks	• Provide tools for managing network resources

Network Slicing Management in Edge and Fog

Model Driven

- Automation and efficient provisioning
- Resource requirements,
- QoS parameters, and
- Service dependencies

Service Centric

- Tailoring network resources to specific services
- Each service receives the necessary resources and guarantees

Network Slicing Management in Edge and Fog

NFV MANO or OpenFog Consortium

- Interoperability and collaboration among different fog computing technologies
- Framework for managing and orchestrating virtualized network functions

Two Layer Abstraction

- facilitates the integration of diverse components in the edge and fog ecosystem
- Abstracting away complexities of underlying hardware and software

Network Slicing Management in Edge and Fog

SDN Based Architecture

- SDN enables programmatic control of network resources
- Dynamic creation, modification, and management of network slices
- Orchestrate the migration and replication of services across the fog and cloud,

Centralized Control

- Centralized view of the network
- Efficient resource allocation and optimization across the entire network

Network Slicing Management in Edge and Fog

Overlay Network

- Isolation of virtual networks
- isolating traffic flows for different services, ensuring that each slice operates independently.

Centralized Control

- Centralized view of the network
- efficient resource allocation and optimization across the entire network

Network Slicing Management in Edge and Fog

- Fog operating system architecture: Designed for IoT services, addressing four main challenges:
  - Scalability
  - Complex inter-networking
  - Dynamic topology and QoS requirements
  - Diversity and heterogeneity in communications and computing

Network Slicing Management in Edge and Fog

- Four main components:
  - Service and device abstraction
  - Resource management
  - Application management
  - Edge resource management

Network Slicing Management in Edge and Fog

- State of Art Solutions Proposed:
  - Lingen et al. [52]: Proposed a model-driven and service-centric architecture for integrating NFV, fog, and 5G/MEC.
  - Truong et al. [53]: Proposed an SDN-based architecture to support fog computing, with use cases in vehicular adhoc networks (VANETs).
  - Bruschi et al. [54]: Proposed a network slicing scheme for supporting multidomain fog/cloud services using SDN-based network slicing.
  - Choi et al. [55]: Proposed a fog operating system architecture called FogOS for IoT services, addressing scalability, inter-networking, dynamics, and diversity challenges.
  - Diro et al. [56]: Proposed a mixed SDN and fog architecture for prioritizing critical network flows while ensuring fairness among other flows.

Network Slicing Management in Edge and Fog

- **Model-driven and service-centric architecture**: Addresses technical challenges of integrating NFV, fog, and 5G/MEC.
- **Open architecture**: Based on NFV MANO (ETSI) and aligned with OpenFog Consortium (OFC) reference architecture.
- **Two-layer abstraction model**: Integrates cloud, network, and fog with IoT-specific modules and enhanced NFV MANO architecture.
- **SDN-based architecture**: Supports fog computing with identified required components and specified roles.
- **Central network view**: SDN controller manages resources and services, optimizes migration and replication.

Network Slicing Management in Edge and Fog

- Use Cases
  - **Smart Cities**: Fog computing can be used for real-time processing of sensor data from smart city infrastructure, such as traffic management and public safety.
  - **Industrial Automation**: Fog computing can be used for real-time control and monitoring of industrial automation systems, such as manufacturing and process control.
  - **Vehicular Networks**: Fog computing can be used for real-time processing of data from vehicular networks, such as traffic management and safety applications.
  - **IoT Applications**: Fog computing can be used for real-time processing of data from IoT devices, such as smart homes and wearables.

Network Slicing Management in Edge and Fog

- **Network slicing scheme**: Supports multidomain fog/cloud services using SDN-based network slicing.
- **Overlay network**: Built using non-overlapping OpenFlow rules for geographically distributed Internet services.
- **Experimental results**: Significant reduction in unicast forwarding rules compared to fully meshed and OpenStack cases.

Network Slicing Management in Edge and Fog

- **Mixed SDN and fog architecture**: Prioritizes critical network flows while ensuring fairness among other flows.

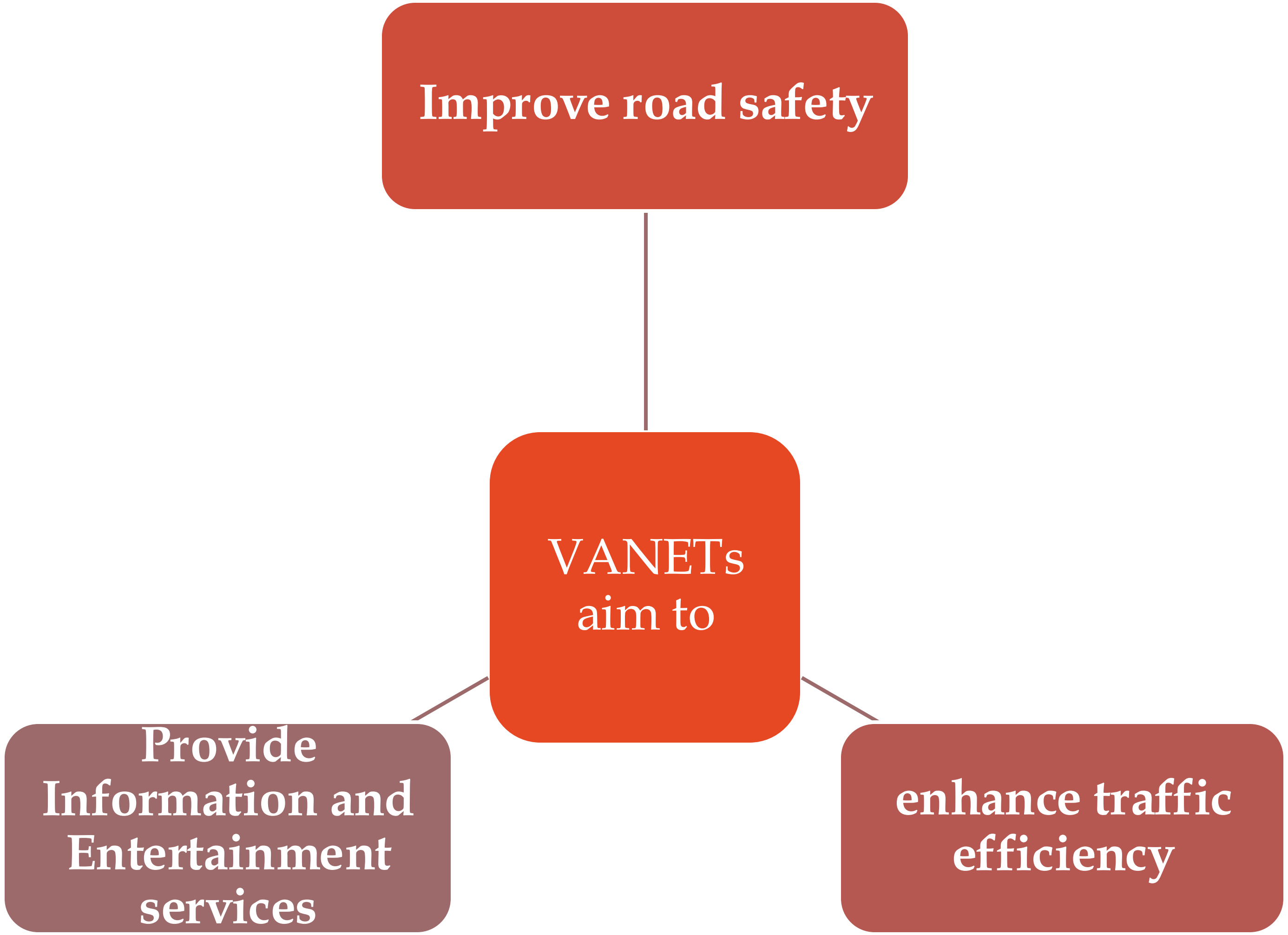
- **QoS requirements**: Satisfies packet delay, lost packets, and maximized throughput for heterogeneous IoT applications.

- **Results**: Efficient serving of critical and urgent flows, with fair allocation of network slices to other flow classes.

Vehicular Ad-hoc Networks (VANETs)

- A **communication system** that enables **vehicles to communicate** with each other (**Vehicle-to-Vehicle or V2V communication**) and with roadside infrastructure (**Vehicle-to-Roadside or V2R communication**).

Vehicular Ad-hoc Networks (VANETs)

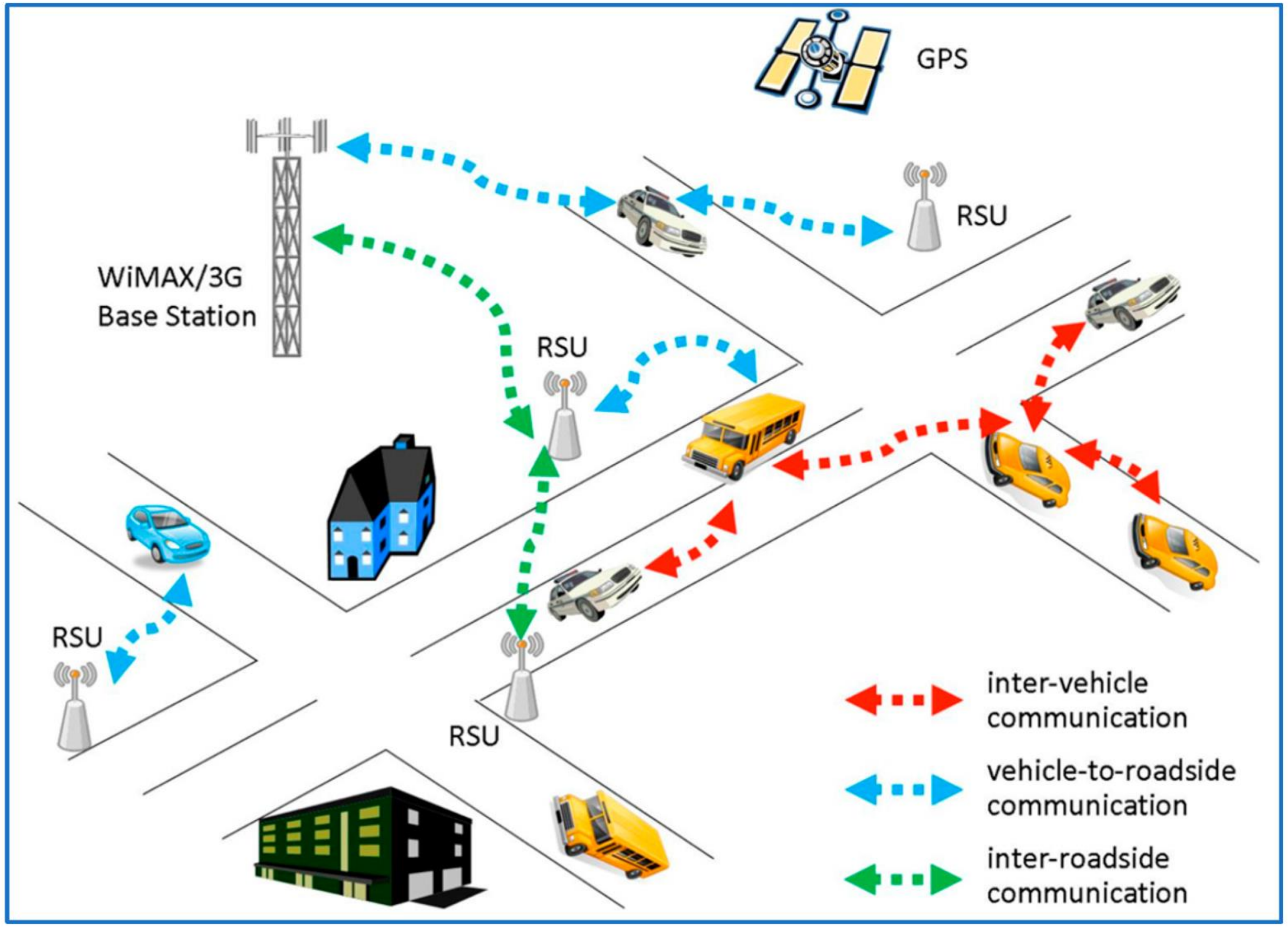


Vehicular Ad-hoc Networks (VANETs)- Conventional

- **Components**
  - **On-board Units**: Vehicles are equipped with various devices such as
    - Sensors
    - Cameras
    - GPS
    - Communication systems to facilitate vehicular communication aimed at enhancing safety and comfort.

Vehicular Ad-hoc Networks (VANETs)- Conventional

➢Communication Architecture

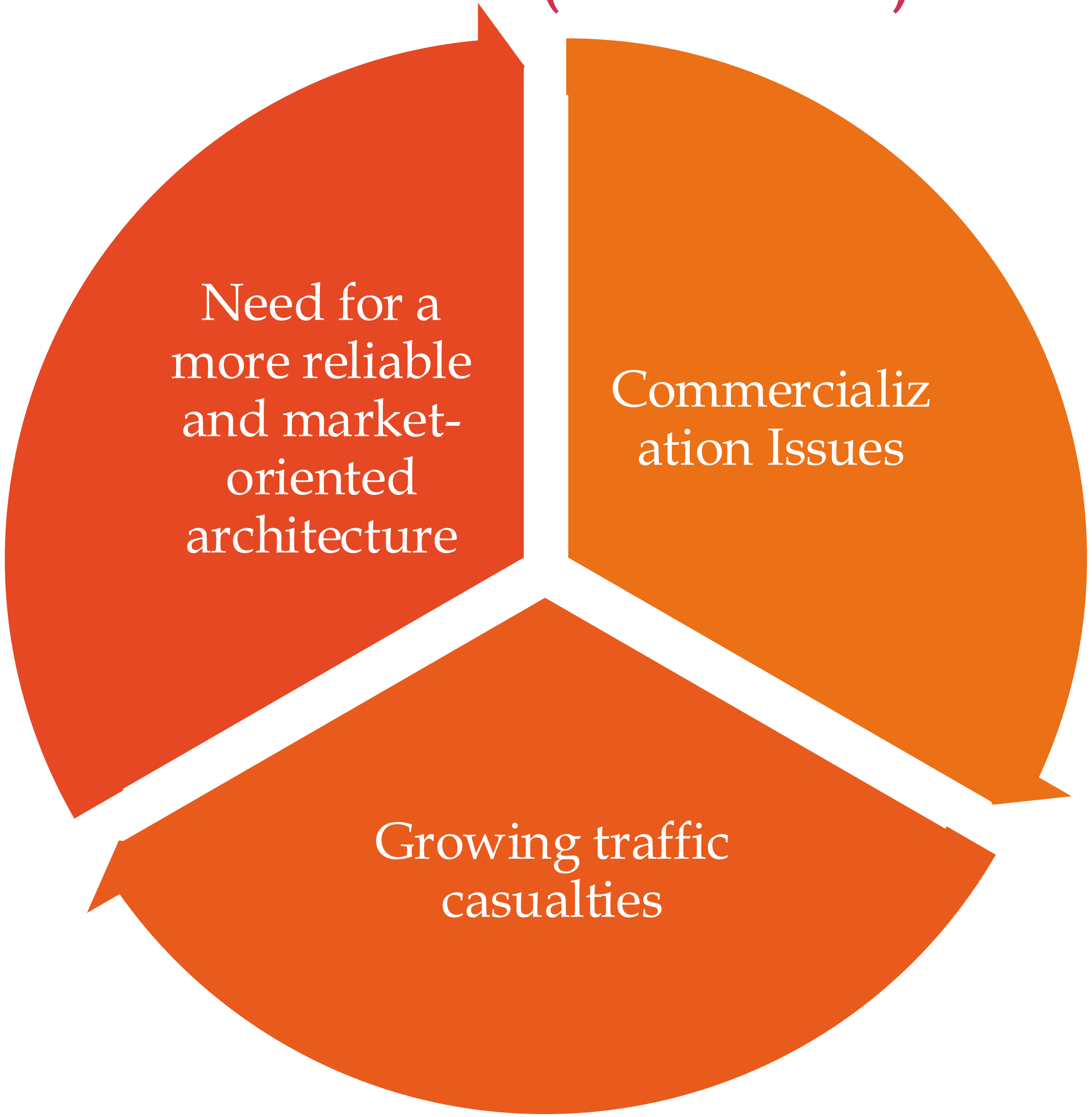


Vehicular Ad-hoc Networks (VANETs)- Conventional

- **Communication Architecture**
  - **WAVE-based Wi-Fi**: In this architecture, Roadside Units (RSUs) act as wireless access points, providing communication coverage to vehicles within their range.
  - **Adhoc Networks**: Vehicles form ad hoc networks using **Wireless Access in Vehicular Environments** (WAVE) technology, enabling them to communicate directly with each other without relying on any fixed infrastructure.
  - **Hybrid Architecture**: This architecture combines cellular and ad hoc architectures using WAVE, allowing vehicles to communicate with each other and with roadside infrastructure using both cellular and ad hoc communication channels..

Vehicular Ad-hoc Networks (VANETs)- Conventional

Limitations



Vehicular Ad-hoc Networks (VANETs)- Conventional

- **Commercialization Issues in VANETs:** Despite their potential, VANETs have not attracted significant commercial interest due to:
  - Inability to provide consistent services due to their adhoc nature.
  - Lack of guaranteed internet connectivity affecting commercial applications.
  - Compatibility issues with personal devices.
  - Limitations in processing capabilities for intelligent decision-making.
  - Lower accuracy in service delivery based on local traffic knowledge.
  - Dependency on user cooperation, which affects reliability.

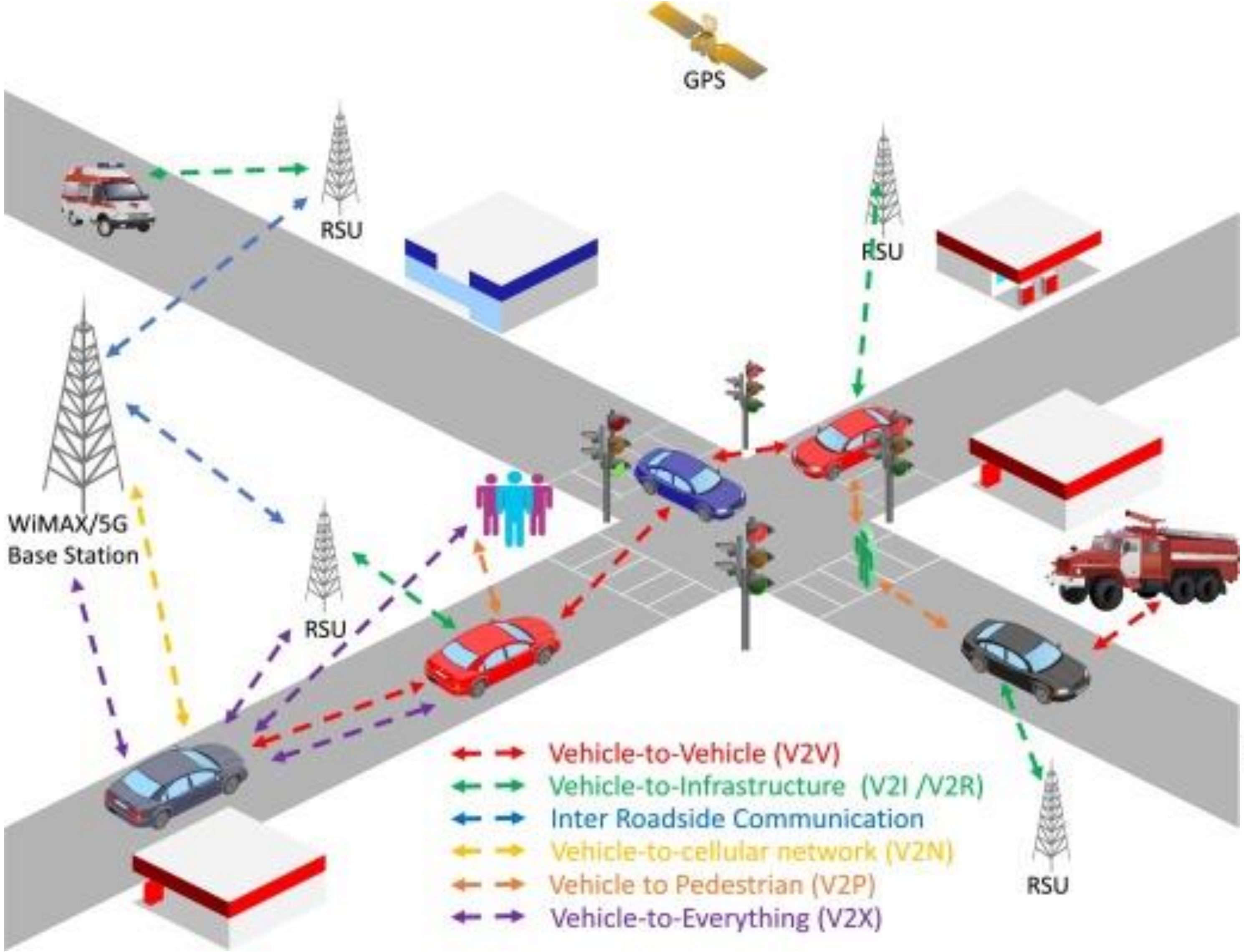
Vehicular Ad-hoc Networks (VANETs)- Conventional

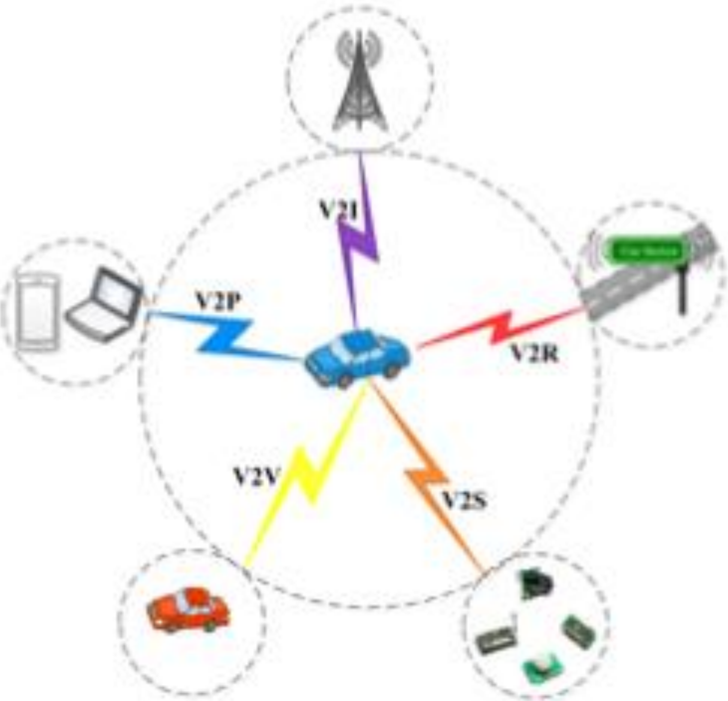
- **Growing Traffic Casualties:**
  - The increasing number of traffic accidents highlights the need for reliable vehicular communication systems.
  - Reports indicate that approximately 1.25 million people die annually from road traffic accidents, emphasizing the urgent need for improved safety measures through IoV.

Introducing Internet of Vehicles (IoV)

- To overcome the limitations of VANETs
- Emerging concept that integrates vehicles into a global network, enabling enhanced communication and data exchange among various entities involved in transportation.
- This network leverages existing technologies to create a more interconnected and intelligent transportation system.

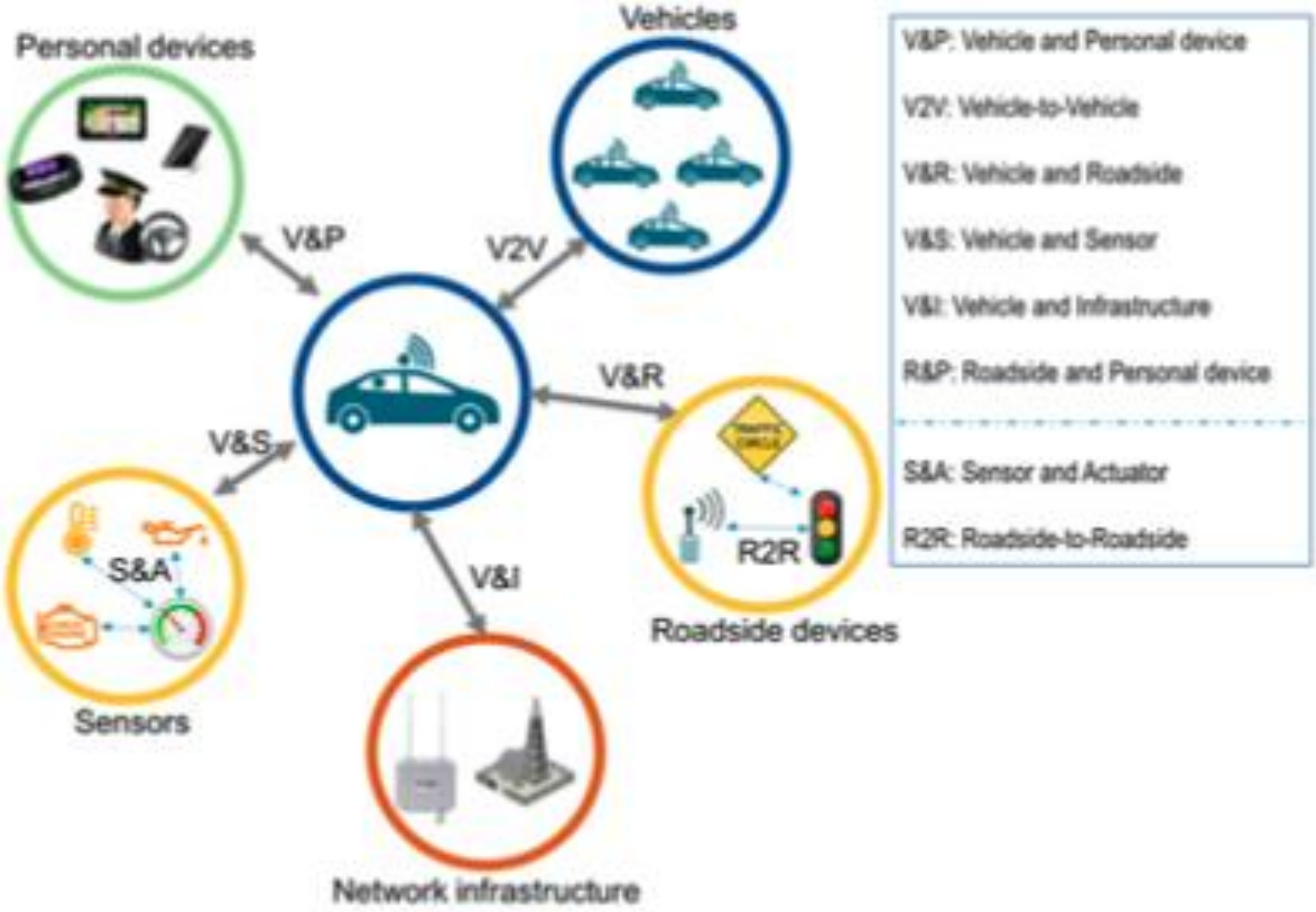
Introducing Internet of Vehicles (IoV)





The five types of vehicular communications of IoV.

Introducing Internet of Vehicles (IoV)



Introducing Internet of Vehicles (IoV) - Ecosystem

- Components
  - Vehicle (V): Nearby vehicles that communicate to exchange information.
  - Person (P): Individuals requesting or accessing services in the IoV environment.
  - Personal Device (P): Devices belonging to individuals that provide or use services.
  - Network Infrastructure (I): Devices facilitating data transmission across the network.
  - Sensing Device (S): Sensors collecting information about vehicles, health levels of individuals, and environmental conditions.
  - Roadside Device (R): Infrastructure elements like traffic lights and information screens that disseminate relevant traffic information.

Introducing Internet of Vehicles (IoV) - Ecosystem

- **Interaction Types:**
- The framework defines several types of Device-to-Device (D2D) interactions:
  - Vehicle-to-Vehicle (V2V)
  - Vehicle-to-Personal (V&P)
  - Vehicle-to-Roadside (V&R)
  - Vehicle-to-Sensor (V&S)
  - Vehicle-to-Infrastructure (V&I)
  - Roadside-to-Personal Device (R&P)
  - Additional internal interactions like Roadside-to-Roadside (R2R) and Sensor & Actuator (S&A).

Introducing Internet of Vehicles (IoV)

- 5 Types of Communication
  - Vehicle-to-Vehicle (V2V)
  - Vehicle-to-Roadside Unit (V2R)
  - Vehicle-to-Infrastructure (V2I)
  - Vehicle-to-Personal Devices (V2P)
  - Vehicle-to-Sensors (V2S)

- Each type utilizes different wireless access technologies (WAT), such as IEEE WAVE for V2V and V2R, Wi-Fi and 4G/LTE for V2I, and CarPlay for V2P.

Internet of Vehicles (IoV) – Layered Architecture

- Creating a universal network framework that includes diverse networks is complex.
- It requires careful identification and grouping of similar functionalities into distinct layers while optimizing the number of layers and enhancing their differentiability.
- Key network characteristics such as interoperability, scalability, reliability, and modularity are essential considerations.

Internet of Vehicles (IoV) – Layered Architecture

- The IoV architecture is designed to be open and adaptable to different technologies.
- It aims for strong integration with the Internet, Service-Oriented Architecture (SOA), and plug-and-play interfaces.
- This flexibility is crucial for accommodating the evolving landscape of vehicular technologies.

Internet of Vehicles (IoV) – Layered Architecture

Layers	Representation	Functionalities
Business	Graphs, Flowchart, Table, Diagram	• Business model and investment designs • Resource usage and application pricing • Budget preparation, data aggregation
Application	Smart applications for vehicles and vehicular dynamics	• Smart, intelligent services to end users • Service discovery and integration • Application usage data and statistics
Artificial Intelligence	Cloud computing, big data analysis, expert systems	• Storing, processing, analysis of data • Analysis based decision making • Service management based on profit
Coordination	Heterogeneous Networks: WAVE, WiFi, LTE	• Unified structure transformation • Interoperability provisions • Secure transportation of information
Perception	Sensor and actuator of vehicles, RSU, personal devices	• Data gathering: vehicle, traffic, devices • Digitization and transmission • Energy optimization at lower layers

Internet of Vehicles (IoV) – Layered Architecture

Perception Layer:

- Comprises sensors and actuators in vehicles, roadside units (RSUs), and personal devices.
- Responsible for gathering data on vehicle status, traffic conditions, and environmental factors.
- Ensures secure transmission of data to the coordination layer.

Coordination Layer:

- Acts as a universal network coordination module that integrates various communication technologies (e.g., WAVE, Wi-Fi, 4G/LTE).
- Processes incoming data from the perception layer, ensuring it is formatted uniformly for further analysis.

Internet of Vehicles (IoV) – Layered Architecture

Artificial Intelligence (AI) Layer:

- Serves as the central processing hub, utilizing cloud infrastructure for data storage, analysis, and decision-making.
- Employs techniques like Vehicular Cloud Computing (VCC) and Big Data Analysis (BDA) to derive insights from the data received.

Application Layer:

- Hosts smart applications that provide services related to traffic safety, efficiency, infotainment, and utility.
- Responsible for delivering intelligent services based on processed information from the AI layer and facilitating user interaction with these services.

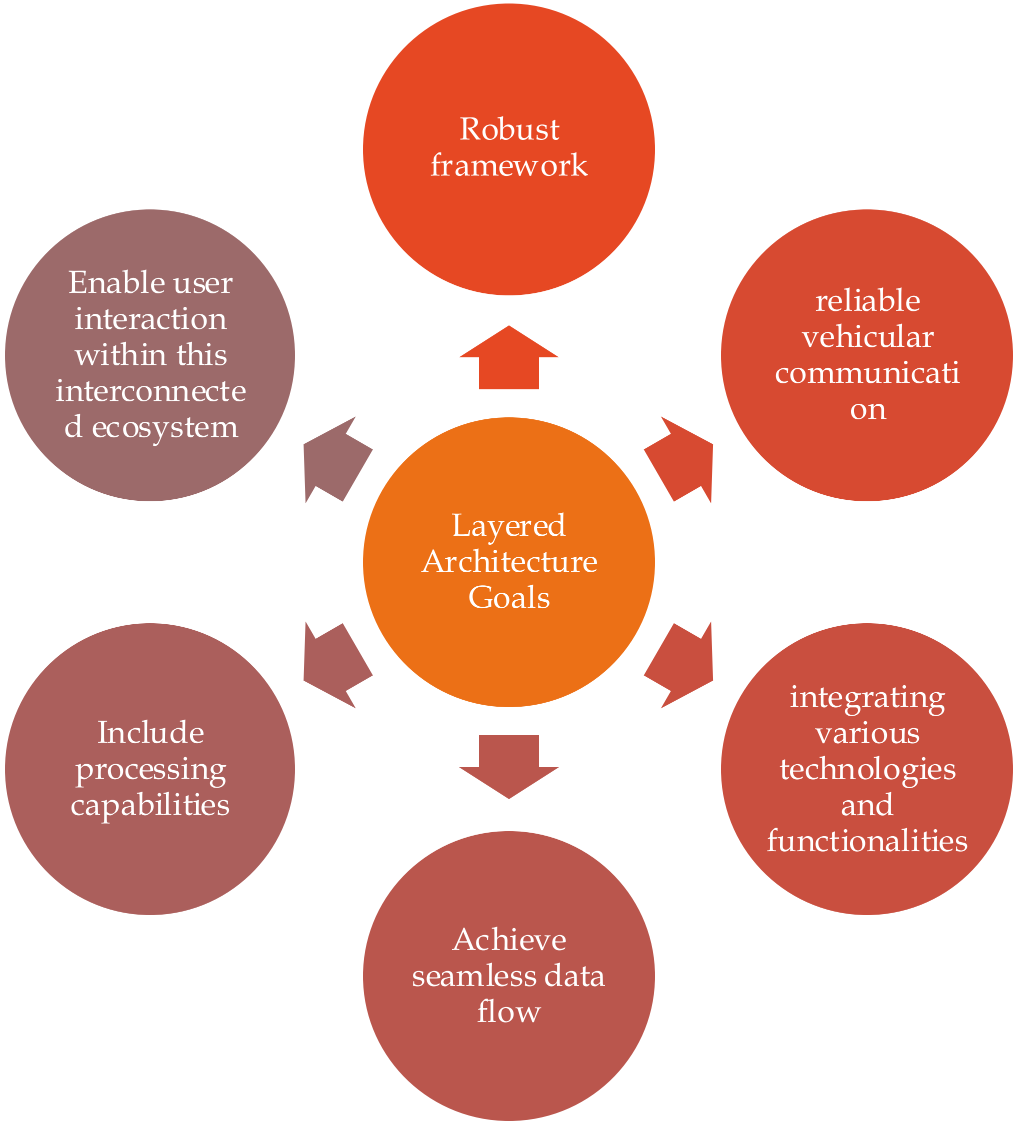
Internet of Vehicles (IoV) – Layered Architecture

Business Layer:

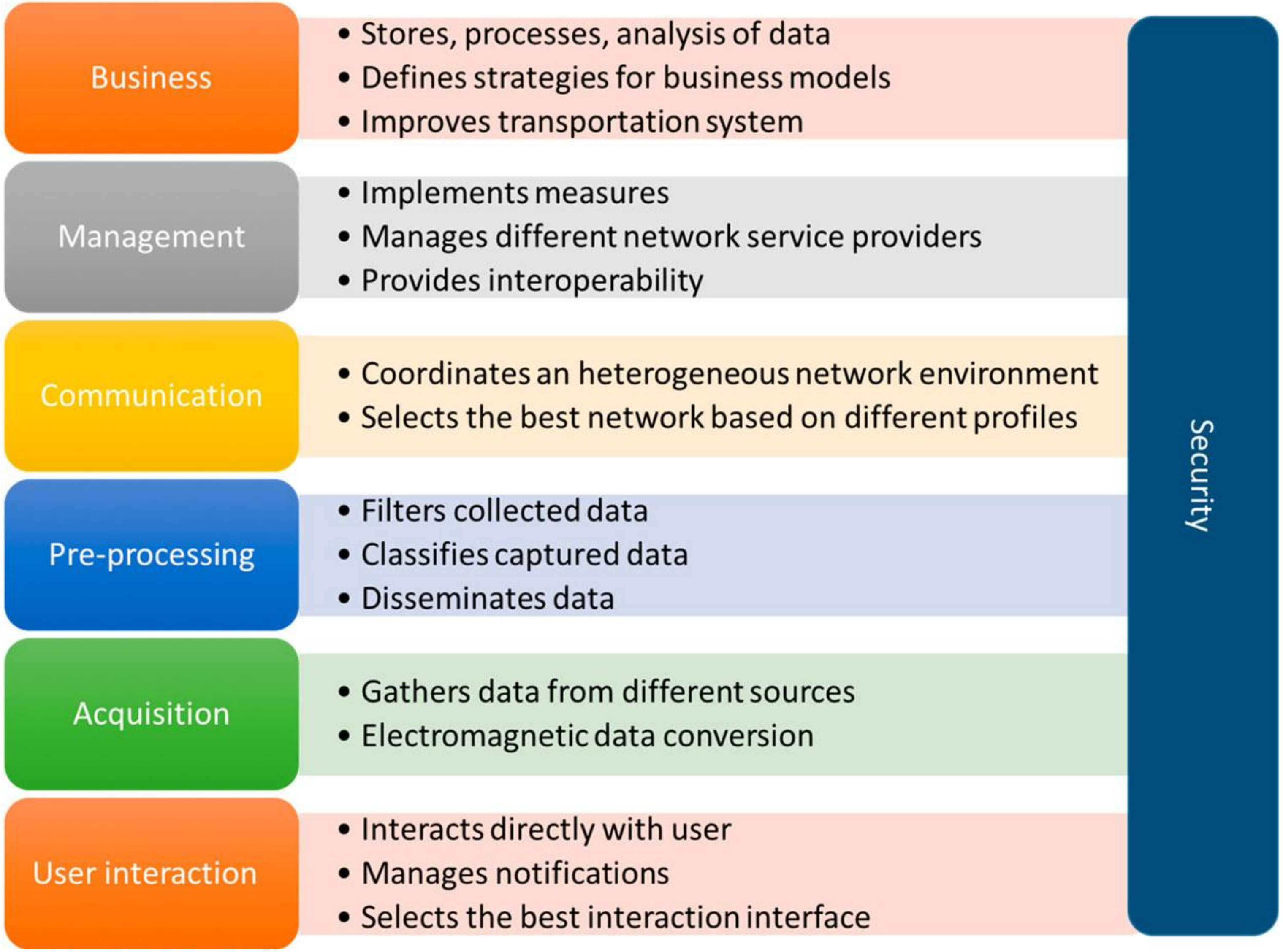
- Focuses on operational management and strategic development of business models based on application usage data.

- Involves decision-making regarding investments, resource allocation, pricing strategies, and overall budget management.

Internet of Vehicles (IoV) – Layered Architecture



Seven Layers security model architecture for IoV



Seven Layers security model architecture for IoV

Layers	Functions	Operations
User Interaction Layer	Provides information-based and control-based systems for drivers.	▪ Displays route information ▪ Traffic conditions, and warnings.
	Enhances driver safety and experience through user-friendly interfaces.	▪ Monitors driving habits and adjusts vehicle controls (e.g., adaptive cruise control).
Data Acquisition Layer	Gathers relevant data from various sources (sensors, GPS, inter-vehicle communication).	Collects data from internal sensors, external devices, and roadside infrastructure.
	Supports intra-vehicular and inter-vehicular interactions.	Facilitates real-time data sharing for safety and infotainment applications

Seven Layers security model architecture for IoV

Layers	Functions	Operations
Data Filtering and Pre-Processing Layer	Analyzes and filters collected data to reduce network traffic.	▪ Implements intelligent data mining techniques to extract relevant information.
	Avoids dissemination of irrelevant information.	▪ Employs techniques for efficient information dissemination among vehicles.
Communication Layer	Integrates multiple wireless networking technologies for seamless connectivity.	Selects optimal networks based on parameters like congestion, QoS, and security.
	Ensures full connectivity for users based on location and service requirements.	Utilizes Multiple Attributes Decision Making (MADM) algorithms for network selection.

Seven Layers security model architecture for IoV

Layers	Functions	Operations
Control and Management Layer	Manages data exchange among various services in the IoV environment.	▪ Implements policies for traffic management, packet inspection, and overall information management.
	Coordinates different network service providers within the IoV ecosystem.	▪ Oversees the flow of information from sensors and roadside infrastructure to users.
Business Layer	Processes large amounts of information using cloud computing infrastructures.	Stores, processes, and analyzes data from other layers for decision-making.
	Develops business strategies based on statistical analysis of data usage.	Provides insights for service providers and government agencies for future infrastructure planning.

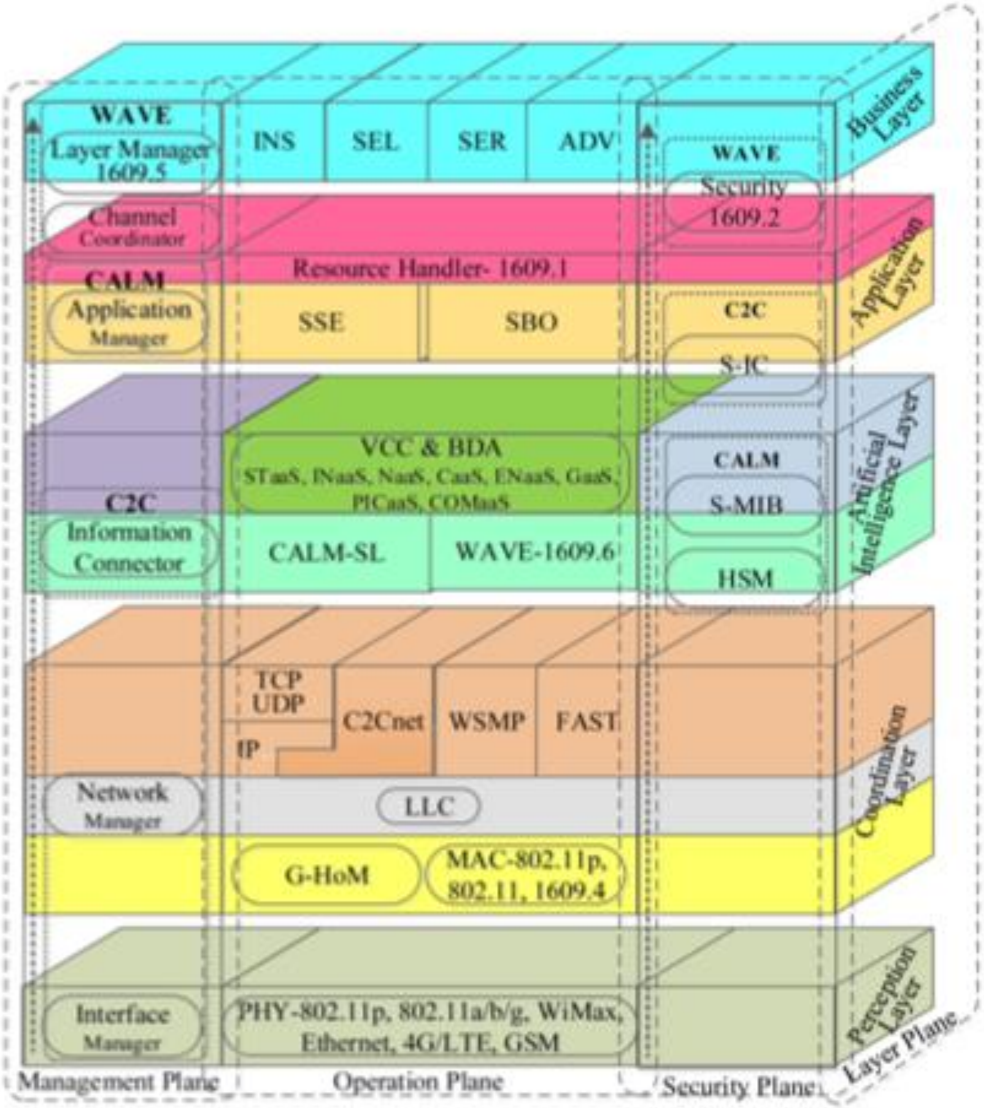
Seven Layers security model architecture for IoV

Layers	Functions	Operations
Security Layer	Implements security measures to protect data at device, network, and application levels.	▪ Ensures data authentication, integrity, confidentiality, access control, and availability in communications.
	Mitigates security threats such as cyberattacks within the IoV ecosystem.	▪ Monitors secure communications among devices, sensors, and actuators through secure networks.

Internet of Vehicles (IoV) – Protocol Stack

- The protocol stack for the Internet of Vehicles (IoV) is a structured framework that organizes various existing protocols across the five layers of the IoV architecture.
- This stack is designed to meet the functional requirements of each layer while ensuring efficient communication and security within the network.

Internet of Vehicles (IoV) – Protocol Stack



Internet of Vehicles (IoV) – Protocol Stack

Security Plane

- This plane focuses on implementing security protocols to protect data and communications within IoV.
  - IEEE 1609.2: A standard for security in vehicular networks.
  - Security Information Connector (S-IC) and Security Management
  - Information Base (S-MIB): Developed under various projects like WAVE, C2C, and CALM.
- Security remains a significant challenge in IoV, as clear definitions for layer-wise security protocols are still being explored.

Internet of Vehicles (IoV) – Protocol Stack

Operation Plane

Perception Layer

Utilizes wireless access protocols such as

- 802.11p (WAVE),
- Wi-Fi standards,
- 4G/LTE, and others for data collection.

Coordination Layer

Divided into three sub-layers:

- **Lower sub-layer:** MAC protocols like IEEE 802.11p.
- **Middle sub-layer:** Logical Link Control (LLC) protocols.
- **Upper sub-layer:** Network layer protocols such as C2C-net and WSMP for data transport.

AI Layer

Uses protocols related to cloud services, including various "as-a-Service" models (e.g., Storage as a Service).

Internet of Vehicles (IoV) – Protocol Stack

Management Plane:

This plane handles management operations within IoV, utilizing protocols from WAVE, CALM, and C2C projects.

Key components include:

- Layer Manager (IEEE 1609.5): For managing channel coordination.
- Application, network, and interface managers proposed by CALM.

Internet of Vehicles (IoV) – Protocol Stack

Plane	Goals	Protocols	Functions
Security Plane	Ensure secure communication and data integrity	IEEE 1609.2	Provides security standards for vehicular networks.
		Security Information Connector (S-IC)	Facilitates secure communication between devices.
		Security Management Information Base (S-MIB)	Manages security policies and configurations.
		Hardware Security Module (HSM)	Provides hardware-based security functions for cryptographic operations.

Internet of Vehicles (IoV) – Protocol Stack

Plane	Goals	Protocols	Functions
Operation Plane	Enable efficient data collection and transmission	802.11p (WAVE)	Wireless access protocol for vehicular communications, enabling V2V and V2R interactions.
		802.11a/b/g (WLAN)	Standards for Wireless Local Area Networks used in various vehicular applications.
		4G/LTE, WiMax, Ethernet	Provides high-speed mobile data communication for vehicles.
		C2C Network Protocol (C2C-net)	Facilitates V2V communication using IPv6.
		WSMP (Wireless Short Message Protocol)	Enables short message communication between vehicles without using IP.
		Fast Application and Communication Enabler (FAST)	Supports application-layer communications without relying on IP.

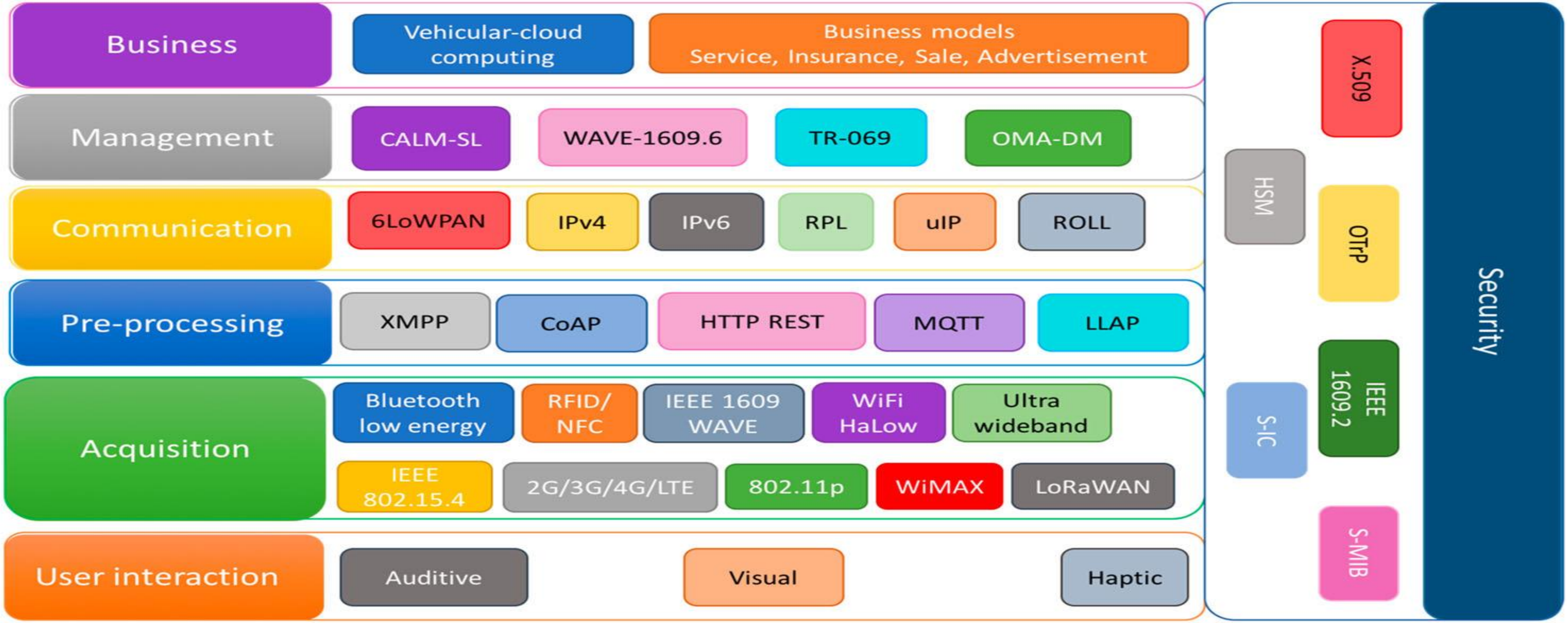
Internet of Vehicles (IoV) – Protocol Stack

Plane	Goals	Protocols	Functions
Operation Plane	Enable efficient data collection and transmission	CALM Service Layer (CALM-SL)	Manages services in a cooperative environment for vehicles.
		Vehicular Cloud Computing (VCC)	Supports cloud-based services for vehicles, enabling data processing and storage.
		Big Data Analysis (BDA)	Analyzes large datasets generated by vehicles for insights and decision-making.

Internet of Vehicles (IoV) – Protocol Stack

Plane	Goals	Protocols	Functions
Management Plane	Manage network operations and resources	IEEE 1609.5	Manages channel coordination in vehicular networks.
		Application Manager Communications Access for Land Mobiles (CALM)	Oversees application management within the network.
		Network Manager (CALM)	Manages network resources and connectivity among vehicles.
		Interface Manager (CALM)	Handles interface interactions between different network components.
		Information Connector Protocol- Car to Car Communication (C2C)	Facilitates information exchange for management purposes in C2C networks.

Internet of Vehicles (IoV) – Protocol Stack



CALM-SL = CALM service layer
OMA-DM = Open Mobile Alliance Device Management
6LoWPAN = IPv6 over Low Power Wireless Personal Area Network
RPL = Routing Protocol for Low Power and Lossy Networks
uIP = micro Internet Protocol
ROLL = Routing Overlow Power and Lossy Networks
XMPP = Xtensinble Messaging and Presents Protocol
CoAP = Constrain Application Protocol

HTTP REST = HypeText Transfer Protocol Representational State Transfer
MQTT = Message Queuing Telemetric Transport
LLAP = Lightweight Local Automation Protocol
LoRaWAN = Low Power Wide Area Network
OTrP = Open Trust Protocol
S-MIB = Security Management Information Base
HSM = Hardware Security Management
S-IC = Security Information Connector

IoV Network Model

- The IoV network model consists of three major elements:
  - Cloud: Represents the brain of IoV, offering cloud services for intelligent computing and processing.
  - Connection: Establishes and maintains communication between the cloud and vehicles.
  - Client: Smart devices, including vehicles, personal devices, and Roadside Units (RSUs), that access cloud services.

IoV Network Model

- **Cloud Element**
- The cloud element has three operation levels:
  - Basic Cloud Services: Offers services like Co-operation as a Service (CaaS), Storage as a Service (STaaS), and Computing as a Service (COaaS).
  - Smart ITS Application Servers: Divided into four categories: traffic safety, traffic management, service subscription, and entertainment.
  - Information Consumer and Producer: Smart devices gather data from vehicular traffic environments.

IoV Network Model

- **Connection Element**
- The connection element enables communication between the cloud and vehicles:
  - Third Party Network Inter Operator (TPNIO): Manages connections, reducing the need for direct Service Level Agreements (SLAs) between network operators.
  - Gateway of Internetworking (GIN): Represents the connection.

IoV Network Model - Connection

- **TPNIO Components -** TPNIO consists of five major components:
  - Global Handoff Manager (GHM): Performs seamless handoffs between network operators.
  - Global Authentication, Authorization, and Billing (GAAB): Verifies vehicle credentials and handles usage-based pricing.
  - Service Management (SM): Provides and monitors quality of service.
  - Network Database (NDB): Stores information on registered networks.
  - Operator Database (ODB): Stores information on registered operators and their SLAs.

IoV Network Model

- **Client Applications -** Client applications in IoV can be broadly divided into two categories:
  - Safety and Management Oriented Clients: These clients focus on traffic safety, navigation, diagnostic, and remote telematics applications.
  - Business Oriented Clients: These clients focus on insurance, car sharing, infotainment, and other business-related applications.

IoV Network Model - Client

- **Safety and Management Oriented Clients** - Some examples of safety and management-oriented clients include:
- Accident Prevention: A M2M communication system that prevents accidents using real-time information exchange between vehicles.
- Emergency Call: An emergency call system that contacts services like police, fire, or family/friends in case of emergency.
- Real-Time Traffic Information: A traffic information system that provides live traffic information using video sensors and heterogeneous communication networks.
- Parking Helper: A parking system that helps find the nearest available parking space by communicating with parked vehicles.
- Self-Repair: A cloud-based step-by-step repair guidance system that helps vehicle owners fix hardware/software issues.

IoV Network Model

- **Business Oriented Clients**

Some examples of business-oriented clients include:

- Insurance on Driving Statistics: An insurance system that automatically calculates insurance fees based on driving statistics.
- Car Pooling: A car sharing application that allocates car service seekers to car owners based on optimization of matching criteria.
- Connected Driving: A device synchronization system that connects a vehicle's display unit to office/home computers, smartphones, or other online devices.
- Cloud Service: A cloud system that forms an autonomous cloud of group vehicles or connects vehicles to traditional clouds, making resources available for usage as cloud services.

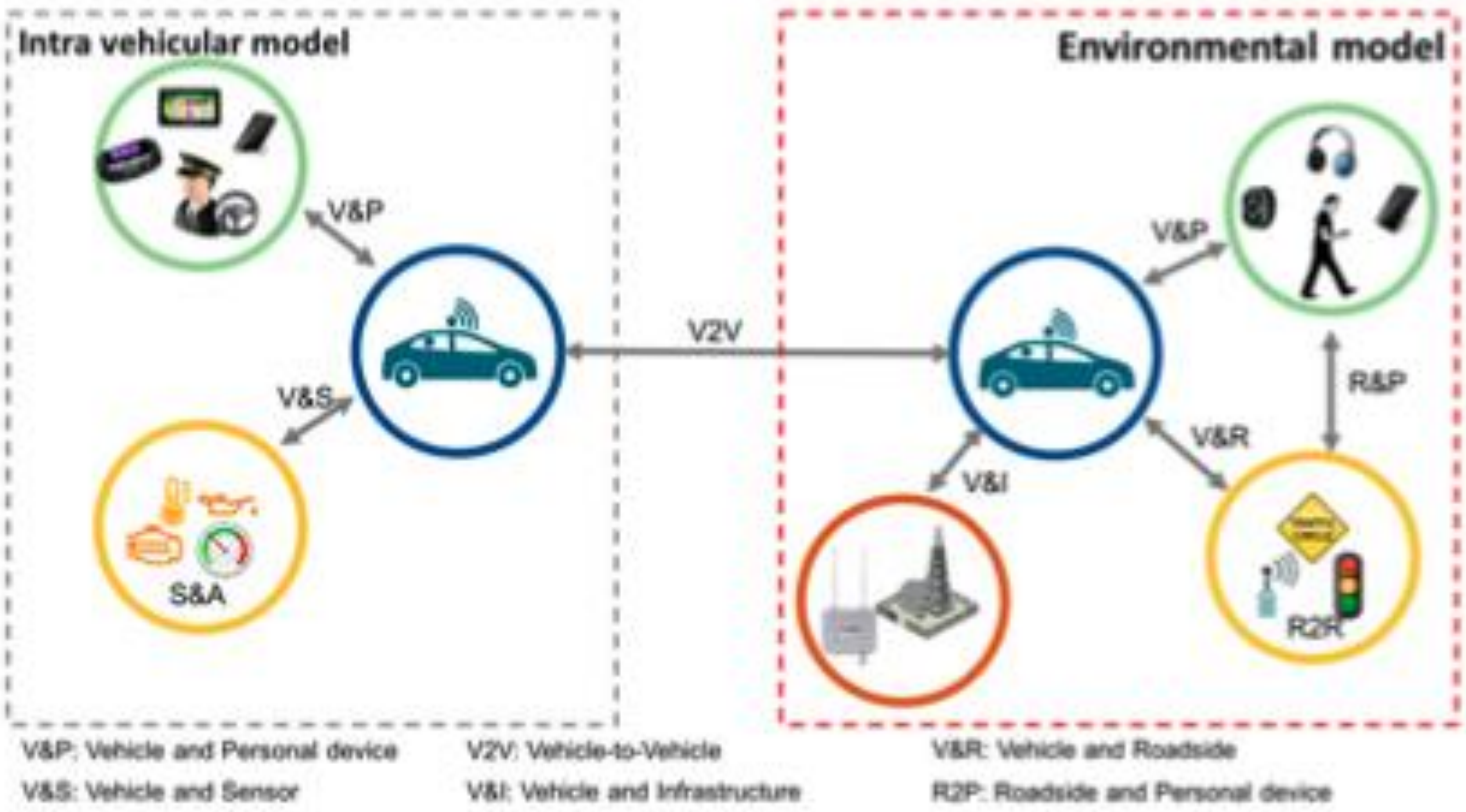
IoV Network Model

- **Prioritized Preference for Clients**

- A prioritized preference of Wireless Access Technologies (WAT) for each type of client is obtained by considering the service requirements of each client in terms of network parameters.
- This prioritized preference would be used by clients to select appropriate WAT while establishing connections.

IoV Network Model

- The Internet of Vehicles (IoV) network model emphasizes a
  - Multi-level collaboration approach that integrates various components and
  - Technologies to enhance vehicle communication and
  - Overall traffic management.



IoV Network Model

Multi-Level Collaboration:

- The IoV network model incorporates multiple users (drivers, passengers, pedestrians), vehicles (cars, motorcycles, bicycles), and devices (sensors, actuators, mobile devices)
- It supports various communication models such as point-to-point, multi-point, broadcast, and geocast.

IoV Network Model

Intra-Vehicular Model:

- Focuses on communication within the vehicle.
- Supports interactions between vehicles and personal devices (V&P), vehicles and sensors (V&S), and sensors and actuators (S&A).

IoV Network Model

Environmental Model:

- Describes communications outside the vehicle.
- Facilitates interactions among multiple vehicles, devices, and networks to improve safety and reduce traffic congestion through various interactions like Vehicle-to-Vehicle (V2V), Vehicle-to-Roadside (V&R), and Vehicle-to-Personal device (V&P).

IoV Network Model - Summary

Elements	Description	Components	Functions
Cloud	Brain of IoV, offers cloud services	Basic Cloud Services, Smart ITS Application Servers, Information Consumer and Producer	Intelligent computing, processing, and storage
Connection	Establishes and maintains communication between cloud and vehicles	Third Party Network Inter Operator (TPNIO), Gateway of Internetworking (GIN)	Mobility management, authentication, authorization, billing
Client	Smart devices that access cloud services	Safety and Management Oriented Clients, Business Oriented Clients	Traffic safety, navigation, diagnostic, remote telematics, insurance, car sharing, infotainment

IoV Network Model – Summary - Cloud

Elements	Description	Functions
Basic Cloud Services	Offers services like CaaS, STaaS, COaaS	Computing, storage, and networking
Smart ITS Application Servers	Divided into four categories: traffic safety, traffic management, service subscription, and entertainment	Traffic management, safety, and entertainment
Information Consumer and Producer	Smart devices that gather data from vehicular traffic environments	Data gathering and dissemination

IoV Network Model – Summary - Connection

Elements	Description	Functions
TPNIO	Manages connections, reduces need for direct SLAs	Mobility management, authentication, authorization, billing
GIN	Represents connection, coordinates with TPNIO	Mobility management, local authentication, authorization, billing
MM	Provides Mobile IP functionalities	Mobility management
LAAB	Provides local authentication, authorization, billing services	Authentication, authorization, billing
TM	Provides network traffic monitoring services	Traffic monitoring

IoV Network Model – Summary – Client App

Client App	Description	Functions
Accident Prevention	M2M communication system	Accident prevention
Emergency Call	Emergency call system	Emergency response
Real-Time Traffic Information	Traffic information system	Traffic monitoring
Parking Helper	Parking system	Parking assistance
Self-Repair	Cloud-based repair guidance system	Vehicle maintenance
Insurance on Driving Statistics	Insurance system	Insurance calculation
Carpooling	Car sharing application	Car sharing
Connected Driving	Device synchronization system	Infotainment
Cloud Service	Cloud system	Cloud computing

IoV Network Model – Summary – Preference

Client Type	Prioritized Preference of WAT
Safety and Management Oriented Clients	1. DSRC (Dedicated Short-Range Communications) 2. Wi-Fi, 3. 4G/LTE (Long-Term Evolution)
Business Oriented Clients	1. 4G/LTE 2. Wi-Fi 3. DSRC

IoV – Key Challenges

- **Localization Accuracy:**
  - Achieving precise vehicle localization is difficult, as current GPS systems provide accuracy of about 5 meters, while IoV requires accuracy within 50 centimeters.
  - Existing GPS systems do not account for vehicle speed, which is essential for effective communication.
  - GPS signals can be unreliable or unavailable in dense urban areas.

- **Location Privacy:**
  - Vehicles frequently share location data, which raises significant privacy concerns.
  - Techniques such as pseudonym switching, silent periods, and mix zones have been proposed to enhance privacy but have limitations:
    - Pseudonym switching is less effective in low-density traffic.
    - Silent periods are not suitable for real-time applications.
    - Mix zones are ineffective on one-way roads.

IoV – Key Challenges

- **Location Verification:**
  - Verifying the location of neighboring vehicles is challenging due to the lack of a trusted authority in vehicular communications.
  - Proposed techniques (like directional antennas and cooperative approaches) face issues such as high infrastructure costs and untrustworthy neighbors.
- **Radio Propagation Model:**
  - Radio signal quality deteriorates in vehicular environments due to moving and static obstacles (e.g., other vehicles, buildings).
  - Current wireless propagation models do not adequately account for these obstacles, leading to unreliable communication.
  - The use of the 5.9 GHz frequency for vehicular communication has limited penetration capabilities.

IoV – Key Challenges

- **Operational Management:**
  - The increase in data volume from numerous sensors and devices complicates operational management in terms of security and reliability.
  - A multi-attribute network will emerge, requiring collaboration among various network operators (ISPs, telecom companies, etc.), which adds complexity.
- **Additional Issues:**
  - Other challenges include disruption reduction, opportunistic frameworks, geographic routing, and MAC standards, which require further investigation within the IoV context.

IoV – Key Future Aspects

- Online Vehicle: Vehicles will be connected online from the moment they are manufactured, allowing access to services like vehicle status, inspection reports, and service history, which will significantly reduce management costs.
- Global Internet ID: Each vehicle will have a unique identifier on the internet, improving accountability in accidents and preventing issues like falsified registrations. This ID will facilitate services similar to a black box for vehicles.
- RFID + GPS Integration: Combining RFID for secure identification with GPS for real-time positioning will enhance the capabilities of Intelligent Transportation Systems (ITS), leading to better route discovery and improved overall performance.
- On-Road Internet: Reliable internet services in vehicles will connect all on-road vehicles to cyberspace, prompting new research into privacy protection and trusted identification systems.

IoV – Key Future Aspects

- **Big Business Data:** The integration of cloud computing with vehicular networks will create vast data resources that can be leveraged across various industries, including automotive and insurance, but will also pose challenges in data management.
- **Smart Terminals:** Demand for machine-to-machine communication terminals will rise, enabling location-based services through collaboration with IoV systems.
- **Car Payment Systems:** Vehicles will use their unique online identities for payments related to tolls, fuel, parking, and other services, streamlining transactions and enhancing traffic management efficiency.